

A Squared-Complex Framework for Square-Prefix Möbius Sums

Exact arithmetic, cofactor geometry, signed Gram structure,
an RH-equivalent local criterion, and Lean formalization

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Abstract

This paper develops a geometric and formal framework for square-prefix values of the Mertens function. At the complete cutoff

$$X_n = (n + 1)^2 - 1,$$

a squarefree integer cannot contain two prime factors larger than $n + 1$. This elementary observation gives an exact smooth-transport decomposition

$$S_n = M(X_n) = A_n - T_n,$$

where A_n is the signed mass already smooth at the square-root cutoff and T_n records sources with one prime factor still beyond it.

The arithmetic is then placed in squared complex coordinates. For a factor pair $cq = m$, write

$$a = \frac{c + q}{2}, \quad b = \frac{q - c}{2}, \quad (a + bi)^2 = m + i \frac{q^2 - c^2}{2}.$$

The real coordinate is the source integer, while the imaginary coordinate

$$Y = 2ab = \frac{q^2 - c^2}{2}$$

records factor imbalance. Fixed cofactors trace explicit parabolas, the prefix cutoff is a moving vertical boundary, and the smoothness cutoff is a moving parabola. Their intersection occurs exactly at the arithmetic transition $q = R$. In these coordinates, a source enters the prefix, occupies the transport region, and later moves into the smooth region; that internal transfer telescopes identically in $A - T$.

The same coordinate Y controls both geometry and time. Low height forces a small factor gap, bounded occupancy in each square block, and bounded transport lifetime. This yields a proved cubic bound for the low-height sector without requiring sign cancellation. The quadratic map is conformal, with Jacobian $c^2 + q^2$, and therefore supplies the two powers of geometric dilution suggested by the cubic energy scale. This scaling mechanism does not by itself prove cancellation; it identifies the scale on which the unresolved high-height interaction must be controlled.

The high sector is formulated as one signed Gram problem rather than a collection of positive shell estimates. Exact dyadic Möbius compression, deterministic prime-3 activation, the congruences $q^2 \equiv 1 \pmod{24}$ and $q^2 \equiv 1 \text{ or } 9 \pmod{40}$, and the corrected modulus- $2r$ quadratic

phase expose the resonant modes that must be separated from the nonresonant packet state. Numerical diagnostics show that individual shell energies can be large even when their signed recombination is small, confirming that the off-diagonal Gram terms are structurally essential.

The Riemann Hypothesis enters through the uniform local square-prefix criterion

$$\sum_{n=N}^{N+H-1} |S_n|^2 \ll_{\varepsilon} H N^{2+\varepsilon}, \quad 1 \leq H \leq N.$$

For every fixed height cutoff Λ , the exact signal decomposition $S_n = S_n^{\text{low}} + S_n^{\text{high}}$ and the elementary bound $|S_n^{\text{low}}| \ll_{\Lambda} n$ imply that this total criterion is *equivalent* to the same uniform local estimate for the high sector alone. Consequently RH is equivalent to the local high-sector signed-energy criterion in the squared-complex geometry; the open analytic problem is to prove that criterion, not to justify the reduction.

The accompanying Lean project imports thirty-nine theorem modules. It formalizes the concrete Mertens summatory function, the exact endpoint $X_n = (n+1)^2 - 1$, interpolation between arbitrary and square-prefix arguments, the pointwise/local conversion, and the total/high geometric equivalence. Its final integration theorem accepts directly the standard external proposition “Mertens criterion $\iff$ RH,” with no project-specific start-sequence, indexing, exponent, localization, or RH-bridge adapter remaining. No project axiom or unconditional proof of RH is claimed.

How to read this paper

The paper has four layers. First, the square-root cutoff gives an exact arithmetic decomposition. Second, squared complex coordinates turn that decomposition into a moving geometric partition with explicit cofactor curves, entry times, and transition times. Third, the remaining high-height problem is organized as a signed Gram and block-contraction problem, supported by reproducible numerical diagnostics. Fourth, the Lean development records which algebraic and geometric layers are machine checked and isolates the analytic statements that still require proof.

Readers interested mainly in the RH connection may begin with [Sections 1, 19, 21 and 24](#). Readers interested in the geometry may begin with [Sections 3, 4, 6 and 7](#). The numerical section is evidence and diagnosis, not a substitute for the missing uniform theorem.

1 Overview and main results

Why this section is here. The choice of square prefixes is not cosmetic. It gives a cutoff with an exact square root, which turns a complicated smoothness condition into a clean one-large-prime decomposition. The squared complex coordinate is then chosen because it records the product and factor imbalance at the same time.

Set

$$X_n = (n+1)^2 - 1, \quad S_n = M(X_n).$$

Why stop immediately before a square? Because if $m < R^2$ is squarefree, it cannot contain two prime factors larger than R . Every source is therefore either already R -smooth or has a unique representation

$$m = cq, \quad c < R < q,$$

with q prime. This gives the exact identity

$$S_n = A_n - T_n.$$

The identity is elementary; the difficult question is why the two much larger terms A_n and T_n track each other so closely.

The geometry is designed to make that question visible. For the factor pair $cq = m$, define

$$a = \frac{c+q}{2}, \quad b = \frac{q-c}{2}.$$

Then

$$(a+bi)^2 = m + i \frac{q^2 - c^2}{2}.$$

The real coordinate remembers the integer m , while the imaginary coordinate measures how uneven the factor pair is. Fixed cofactors become parabolas. The complete prefix is cut out by a vertical line. Smoothness is cut out by a second moving parabola. A source first enters the prefix, then spends time in the transport region, and finally becomes smooth when the moving boundary reaches its largest prime factor.

This viewpoint separates three logically different mechanisms:

1. *exact bookkeeping*: the transport-to-smooth crossing cancels identically in $A - T$;
2. *geometric dilution*: the quadratic map expands area by order n^2 at square scale n ;
3. *analytic cancellation*: the remaining high-height contribution must be controlled through one signed Gram form, including its off-diagonal terms.

The first two are proved here. The third is the central open estimate. The coordinate change does not prove cancellation by itself; it identifies the correct state space, scale, and interaction that a proof must control.

The main exact results are summarized below.

Theorem 1.1 (Exact square-root decomposition). *For $R_n = n + 1$ and $X_n = R_n^2 - 1$, define*

$$A_n := \sum_{\substack{m \leq X_n \\ P^+(m) \leq R_n}} \mu(m),$$

$$T_n := \sum_{c < R_n} \mu(c) \left[\pi\left(\left\lfloor \frac{X_n}{c} \right\rfloor\right) - \pi(R_n) \right].$$

Then

$$\boxed{S_n = A_n - T_n.}$$

Theorem 1.2 (Squared-space recovery). *Let $w = x + iy$ be generated by a positive factor pair $cq = x$ through*

$$y = \frac{q^2 - c^2}{2}.$$

With $\rho = |w|$,

$$\boxed{c^2 = \rho - y, \quad q^2 = \rho + y.}$$

Hence the squared point determines the ordered factor pair exactly.

Theorem 1.3 (Cofactor-curve cutoff intersection). *For fixed $c > 0$, put*

$$\gamma_c(x) = \frac{x^2 - c^4}{2c^2}.$$

At cutoff $R > 0$, put

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}.$$

Then

$$\boxed{\gamma_c(x) = \Gamma_R(x) \iff x = cR.}$$

The curves meet orthogonally at their intersection. At an arithmetic point $x = cq$, the sign of $\gamma_c(cq) - \Gamma_R(cq)$ is the sign of $q - R$.

Theorem 1.4 (Exact channel telescoping). *As R increases, a canonical point enters the prefix at $R = \sqrt{cq}$ and crosses from transport to smooth at $R = q$. The signed mass crossing the moving parabola appears with opposite signs in the increments of A and $-T$. Therefore all internal channel transfers cancel from the increment of $A - T$; only points entering the new square strip remain.*

Theorem 1.5 (Low-height clustering and lifetime). *Let $m = cq \in B_n$ be canonical and let*

$$Y_m = \frac{q^2 - c^2}{2}.$$

Then, for $H \geq 0$,

$$\#\{m \in B_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor,$$

and the count is zero when $H = n$. For odd sources the bound improves to $\lfloor H/(2n) \rfloor$. If $Y_m > 0$, the transport lifetime satisfies

$$L(m) \leq \frac{2Y_m}{q}.$$

Consequently, for fixed Λ , the canonical points with $|Y_m| \leq \Lambda n$ have uniformly bounded block occupancy, and the transport members have uniformly bounded lifetime.

Theorem 1.6 (Major-arc rank-one law for cofactor blocks). *Fix $\eta > 0$ and a dyadic cofactor scale $C \asymp N^\alpha$ with $0 \leq \alpha \leq 1 - \eta$. On the local square-prefix interval $N \leq n < 2N$, let*

$$B_C(n) := - \sum_{C < c \leq 2C} \mu(c) \Pi_c(X_n), \quad a_C := \sum_{C < c \leq 2C} \frac{\mu(c)}{c},$$

and let $P_n = -\pi(X_n)$ be the prime-curve vector. Then

$$\boxed{B_C(n) = \frac{2}{2 - \alpha} a_C P_n + O_\eta \left(\frac{N^2}{\log^2 N} \right)}$$

uniformly for $N \leq n < 2N$. Consequently, the local Gram matrix of any fixed collection of such cofactor blocks is a rank-one prime-curve main term plus an error of order $O_\eta(N^5 / \log^3 N)$ entrywise.

Theorem 1.7 (Quadratic conformal expansion). *The map*

$$\Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right)$$

has

$$D\Phi(c, q)^\top D\Phi(c, q) = (c^2 + q^2)I,$$

and

$$\det D\Phi(c, q) = c^2 + q^2 \geq 2cq.$$

At square scale $cq \asymp n^2$, lengths expand by at least order n and areas by at least order n^2 .

Theorem 1.8 (Exact dyadic cell compression). *For every $k \geq 0$,*

$$\boxed{(\mu(4k+1), \mu(4k+2), \mu(4k+3), \mu(4k+4)) = (\mu(4k+1), -\mu(2k+1), \mu(4k+3), 0)}.$$

Consequently the complete-cell scalar is

$$\boxed{C_k = \mu(4k+1) - \mu(2k+1) + \mu(4k+3)}.$$

Among the three active slots $4k+1, 4k+2, 4k+3$, exactly one is divisible by 3; the affected slot is $4k+3, 4k+2, 4k+1$ according as $k \equiv 0, 1, 2 \pmod{3}$.

Theorem 1.9 (Exact small-modulus quadratic resonance). *For every prime $q > 3$,*

$$\boxed{q^2 \equiv 1 \pmod{24}}.$$

Hence, for every integer m and every prime interval containing no 2 or 3,

$$\boxed{\sum_{\substack{Q < q \leq 2Q \\ q \text{ prime}}} e\left(\frac{mq^2}{24}\right) = e\left(\frac{m}{24}\right) (\pi(2Q) - \pi(Q))}.$$

Equivalently, the quadratic prime phase is perfectly coherent at every $\alpha = m/12$. More generally, for $\alpha = a/r$, the rational phase $e(aq^2/(2r))$ is periodic modulo $2r$; the correct reduced-residue model is therefore formed over $(\mathbb{Z}/2r\mathbb{Z})^\times$.

The remaining theorem is an operator estimate: prove that the high-height, cross-curve signed interactions inherit this geometric dilution when inserted into the exact square-prefix kernel. No claim is made that a coordinate change alone proves cancellation; the point is that the coordinate change exposes the exact two-power scale and localizes the unresolved interaction.

2 Square prefixes and the arithmetic decomposition

Why this section is here. The square-root cutoff is the arithmetic hinge of the paper. Before introducing geometry, we isolate the exact decomposition that the geometry is meant to explain.

Define the complete square block

$$B_n := \{m \in \mathbb{N} : n^2 \leq m < (n+1)^2\}, \quad (1)$$

the square-root cutoff

$$R_n := n + 1, \quad (2)$$

and the complete-prefix endpoint

$$X_n := R_n^2 - 1 = (n+1)^2 - 1. \quad (3)$$

The square-prefix Mertens value is

$$S_n := M(X_n) = \sum_{m \leq X_n} \mu(m). \quad (4)$$

Because $\mu(R_n^2) = 0$ for $R_n > 1$, one may equally write $S_n = M(R_n^2)$.

Write $P^+(m)$ for the largest prime factor of m , with $P^+(1) = 1$.

Definition 2.1 (Smooth and transport terms). Define

$$A_n := \sum_{\substack{1 \leq m \leq X_n \\ P^+(m) \leq R_n}} \mu(m), \quad (5)$$

$$T_n := \sum_{1 \leq c < R_n} \mu(c) \left[\pi\left(\left\lfloor \frac{X_n}{c} \right\rfloor\right) - \pi(R_n) \right]. \quad (6)$$

Lemma 2.2 (One-large-prime closure). *Let $m \leq R^2 - 1$ be squarefree. If $P^+(m) > R$, then there is a unique factorization*

$$m = cq, \quad 1 \leq c < R < q, \quad q \text{ prime}, \quad (7)$$

where $q = P^+(m)$. Moreover,

$$\mu(m) = -\mu(c). \quad (8)$$

Proof. Put $q = P^+(m)$ and $c = m/q$. If $c \geq R$, then $m = cq > R^2$, contrary to $m \leq R^2 - 1$. Since m is squarefree, $q \nmid c$ and $\mu(cq) = -\mu(c)$. Uniqueness follows from the largest-prime choice. $\square$

Proof of Theorem 1.1. Partition the squarefree $m \leq X_n$ according to whether $P^+(m) \leq R_n$ or $P^+(m) > R_n$. The first population contributes A_n . By Lemma 2.2, the second population is uniquely $m = cq$ with $c < R_n < q$, and each source has weight $-\mu(c)$. The total second contribution is therefore $-T_n$. $\square$

2.1 The largest-prime cofactor identity

For $c \geq 1$, define the prime-tail counting function

$$\Pi_c(x) := \#\left\{q \text{ prime} : P^+(c) < q \leq \left\lfloor \frac{x}{c} \right\rfloor\right\}. \quad (9)$$

Equivalently,

$$\Pi_c(x) = \left(\pi\left(\left\lfloor \frac{x}{c} \right\rfloor\right) - \pi(P^+(c))\right)_+, \quad (10)$$

where $(u)_+ := \max\{u, 0\}$. The positive part is essential when $x/c < P^+(c)$.

Theorem 2.3 (Exact largest-prime cofactor expansion). *For every real $x \geq 1$,*

$$M(x) = 1 - \sum_{c \geq 1} \mu(c) \Pi_c(x). \quad (11)$$

The sum is finite, and separating the prime curve $c = 1$ gives

$$M(x) = 1 - \pi(x) - \sum_{2 \leq c \leq x/2} \mu(c) \left(\pi\left(\left\lfloor \frac{x}{c} \right\rfloor\right) - \pi(P^+(c)) \right)_+. \quad (12)$$

Proof. Every squarefree $m > 1$ has a unique largest prime factor $q = P^+(m)$. Writing $c = m/q$, one has

$$q > P^+(c), \quad \mu(m) = \mu(cq) = -\mu(c).$$

Conversely, every pair consisting of a squarefree c and a prime $q > P^+(c)$ produces the squarefree integer $m = cq$ whose largest prime factor is q . The condition $m \leq x$ is exactly $q \leq x/c$. Summing $-\mu(c)$ over these unique pairs proves Equation (11). The $c = 1$ term is $-\pi(x)$, because $P^+(1) = 1$. No term with $c > x/2$ can occur. $\square$

Remark 2.4 (Central analytic engine). Equation (12) is the exact arithmetic form of the cofactor-parabola decomposition. It shows that the Mertens function is not obtained by proving that the prime curve is small. The prime curve $-\pi(x)$ is cancelled by the Möbius-weighted collection of all curves $c \geq 2$. The geometric and large-sieve parts of the paper are methods for controlling the rate of this exact cancellation on square prefixes.

2.2 A proved rank-one major-arc approximation

The first analytic component of the cofactor cascade can be proved unconditionally. It explains why the small-cofactor Gram matrix is nearly rank one and identifies the scalar coefficients governing its large coherent mode.

For a dyadic block $C < c \leq 2C$, define

$$B_C(n) := - \sum_{C < c \leq 2C} \mu(c) \Pi_c(X_n), \quad a_C := \sum_{C < c \leq 2C} \frac{\mu(c)}{c}, \quad (13)$$

and write

$$P_n := -\pi(X_n). \quad (14)$$

Theorem 2.5 (Polynomial-range major-arc rank-one law). *Fix $\eta > 0$. Let N tend to infinity, let $0 \leq \alpha \leq 1 - \eta$, and choose a dyadic C satisfying $C \asymp N^\alpha$. Uniformly for $N \leq n < 2N$,*

$$B_C(n) = \lambda_\alpha a_C P_n + O_\eta \left(\frac{N^2}{\log^2 N} \right), \quad \lambda_\alpha := \frac{2}{2 - \alpha}. \quad (15)$$

Consequently,

$$\sum_{N \leq n < 2N} |B_C(n) - \lambda_\alpha a_C P_n|^2 \ll_\eta \frac{N^5}{\log^4 N}. \quad (16)$$

If $D \asymp N^\beta$ with $0 \leq \beta \leq 1 - \eta$, then

$$\langle B_C, B_D \rangle_{[N, 2N]} = \lambda_\alpha \lambda_\beta a_C a_D \|P\|_{[N, 2N]}^2 + O_\eta \left(\frac{N^5}{\log^3 N} \right). \quad (17)$$

Proof. For $c \asymp N^\alpha$ and $N \leq n < 2N$, one has $X_n/c \gg_\eta N^{1+\eta}$, while $P^+(c) \leq c \ll N^{1-\eta}$. Thus, for sufficiently large N , the positive part in $\Pi_c(X_n)$ is inactive and

$$\Pi_c(X_n) = \pi(X_n/c) - \pi(P^+(c)).$$

The prime number theorem with its standard first remainder term gives, uniformly in this range,

$$\pi(X_n/c) = \frac{X_n/c}{\log(X_n/c)} + O_\eta \left(\frac{X_n}{c \log^2 N} \right),$$

and

$$\pi(X_n) = \frac{X_n}{\log X_n} + O \left(\frac{X_n}{\log^2 N} \right).$$

Since

$$\frac{\log X_n}{\log(X_n/c)} = \frac{2}{2-\alpha} + O_\eta \left(\frac{1}{\log N} \right),$$

it follows that

$$\pi(X_n/c) = \frac{\lambda_\alpha}{c} \pi(X_n) + O_\eta \left(\frac{N^2}{c \log^2 N} \right).$$

Summing the error uses $\sum_{C < c \leq 2C} c^{-1} \ll 1$. The fixed lower-prime exclusions satisfy

$$\sum_{C < c \leq 2C} \pi(P^+(c)) \leq \sum_{C < c \leq 2C} c \ll C^2 \ll_\eta N^{2-2\eta},$$

which is absorbed by $N^2/\log^2 N$. This proves Equation (15). Squaring and summing over N local prefixes gives Equation (16). Finally, write each block as its prime-vector main term plus its error and apply Cauchy–Schwarz, using $\|P\|_{[N, 2N]} \asymp N^{5/2}/\log N$. $\square$

Corollary 2.6 (Rank-one major-arc Gram matrix). *For finitely many polynomial cofactor blocks with exponents bounded by $1 - \eta$, their local Gram matrix has the form*

$$G_N = \|P\|_{[N, 2N]}^2 vv^\top + E_N, \quad v_j = \lambda_{\alpha_j} a_{C_j}, \quad (18)$$

with every entry of E_N bounded by $O_\eta(N^5/\log^3 N)$. Thus the coherent major-arc component cancels between dyadic cofactor blocks entirely through the scalar harmonic Möbius coefficients a_{C_j} , before any minor-arc estimate is used.

Remark 2.7 (What this theorem accomplishes). Theorem 2.5 is a genuine major-arc cancellation theorem for substantial ranges $C \asymp N^\alpha$, $\alpha < 1$. It is not RH: it does not control cofactor scales $C \gtrsim N$, nor does it supply a power saving for the rank-one remainder. It does prove that the large small-cofactor interactions are analytically organized rather than arbitrary, and it identifies the exact main term that the Squared-Cofactor Large Sieve must remove.

For

$$\mathbf{A}_N = (A_1, \dots, A_N)^\top, \quad \mathbf{T}_N = (T_1, \dots, T_N)^\top, \quad \mathbf{S}_N = (S_1, \dots, S_N)^\top,$$

we have

$$\mathbf{S}_N = \mathbf{A}_N - \mathbf{T}_N \tag{19}$$

and

$$\sum_{n \leq N} |S_n|^2 = \|\mathbf{A}_N - \mathbf{T}_N\|_2^2. \tag{20}$$

3 All factor pairs above one integer

Why this section is here. A fixed integer has many factor pairs. This section describes the complete geometric fibre first, so that the later canonical Möbius point is not confused with the whole factorization geometry.

The factorization geometry and the smooth-transport geometry use the same coordinate map but different point selections. This distinction is essential.

Fix $N \geq 1$ and consider every positive factor pair

$$uv = N, \quad 1 \leq u \leq v. \tag{21}$$

Define the Fermat coordinates

$$a = \frac{u+v}{2}, \quad b = \frac{v-u}{2} \geq 0. \tag{22}$$

Then

$$u = a - b, \quad v = a + b, \quad N = a^2 - b^2. \tag{23}$$

The coordinates lie in $\frac{1}{2}\mathbb{Z}$; they are integers when u and v have the same parity.

Squaring the complex factor gives

$$(a + bi)^2 = N + iY, \quad \boxed{Y = 2ab = \frac{v^2 - u^2}{2}}. \tag{24}$$

Thus all factorizations of a fixed N lie on the vertical fibre

$$\operatorname{Re} w = N. \tag{25}$$

3.1 Square-root tests

Let

$$w = N + iY, \quad \rho = |w| = \sqrt{N^2 + Y^2}. \tag{26}$$

Then

$$a^2 = \frac{\rho + N}{2}, \tag{27}$$

$$b^2 = \frac{\rho - N}{2}, \tag{28}$$

$$u^2 = \rho - Y, \quad (29)$$

$$v^2 = \rho + Y. \quad (30)$$

Accordingly, a candidate vertical ordinate Y corresponds to an exact factorization if and only if the relevant square and parity conditions hold. The identity

$$\sqrt{N^2 + Y^2} = a^2 + b^2 \quad (31)$$

is the right-triangle form of the same condition.

3.2 Admissible intervals

For a factor pair $u(N/u) = N$ with $1 \leq u \leq \sqrt{N}$,

$$a(u) = \frac{u + N/u}{2}, \quad b(u) = \frac{N/u - u}{2}. \quad (32)$$

Both functions decrease as u increases toward $\sqrt{N}$. Therefore the full positive factor fibre satisfies

$$\sqrt{N} \leq a \leq \frac{N+1}{2}, \quad (33)$$

$$0 \leq b \leq \frac{N-1}{2}, \quad (34)$$

$$0 \leq Y \leq \frac{N^2-1}{2}. \quad (35)$$

For integer enumeration one uses $a \geq \lceil \sqrt{N} \rceil$ with the parity required by N .

The upper endpoint corresponds to the trivial factor pair $1 \cdot N$. If the trivial factor is excluded and the smallest permitted odd factor is 3, then

$$a \leq \frac{N+9}{6}, \quad (36)$$

$$b \leq \frac{N-9}{6}, \quad (37)$$

$$Y \leq \frac{N^2-81}{18}. \quad (38)$$

These are the nontrivial odd-factor or semiprime-search envelopes. They are not the universal bounds for the complete factor fibre.

Example 3.1 (The fibre $N = 309$). The trivial factor pair $1 \cdot 309$ gives

$$(a, b) = (155, 154), \quad w = 309 + 47740i.$$

The nontrivial factorization $3 \cdot 103$ gives

$$(a, b) = (53, 50), \quad w = 309 + 5300i.$$

The latter is the highest point allowed by the nontrivial odd-factor condition $u \geq 3$, but not the highest point on the full factor fibre.

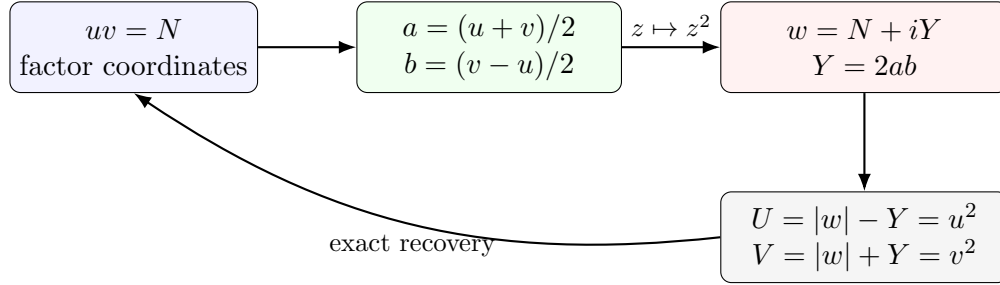


Figure 1: Equivalent coordinate systems on a complete factor fibre. The squared coordinates retain the factor pair exactly.

4 The canonical Möbius geometry

Why this section is here. The Möbius decomposition needs one reproducible point per squarefree source. Choosing the largest prime factor supplies that point and gives a signed height that can be followed across cutoffs.

The smooth-transport decomposition does not use every point on each factor fibre. It selects one canonical point per squarefree source.

For squarefree $m > 1$, define

$$q_m := P^+(m), \quad c_m := \frac{m}{q_m}. \quad (39)$$

Unlike the ordered factor pair in [Section 3](#), the canonical pair is not reordered. The cofactor c_m may exceed q_m . For example,

$$m = 30, \quad q_m = 5, \quad c_m = 6.$$

Therefore define the signed Fermat coordinate

$$a_m := \frac{c_m + q_m}{2}, \quad b_m := \frac{q_m - c_m}{2} \in \frac{1}{2}\mathbb{Z}, \quad (40)$$

and the canonical squared point

$$w_m := (a_m + ib_m)^2 = m + iY_m, \quad (41)$$

where

$$Y_m = 2a_mb_m = \frac{q_m^2 - c_m^2}{2}. \quad (42)$$

The sign of Y_m records the order of the canonical factors:

$$Y_m > 0 \iff c_m < q_m, \quad Y_m < 0 \iff c_m > q_m. \quad (43)$$

4.1 Parameterization by cofactors and primes

Every squarefree $m > 1$ has a unique representation

$$m = cq, \quad (44)$$

where

$$\mu(c) \neq 0, \quad q \text{ is prime}, \quad q > P^+(c). \quad (45)$$

Conversely, every pair satisfying Equation (45) produces a squarefree integer $m = cq$ with largest prime factor q . Moreover,

$$\mu(cq) = -\mu(c). \quad (46)$$

Define the canonical index set

$$\mathcal{C} := \{(c, q) : \mu(c) \neq 0, \quad q \text{ prime}, \quad q > P^+(c)\}. \quad (47)$$

The squared point cloud is

$$\mathcal{W} := \left\{ cq + i \frac{q^2 - c^2}{2} : (c, q) \in \mathcal{C} \right\}. \quad (48)$$

The signed point measure is

$$\nu := \delta_1 - \sum_{(c,q) \in \mathcal{C}} \mu(c) \delta_{cq + i(q^2 - c^2)/2}. \quad (49)$$

The atom at 1 records $\mu(1) = 1$.

Example 4.1 (A point below the real axis). For $m = 30$, the canonical pair is $(c, q) = (6, 5)$, so

$$a = \frac{11}{2}, \quad b = -\frac{1}{2}, \quad w_{30} = 30 - \frac{11}{2}i.$$

This point enters a square prefix only after the cutoff already exceeds $q = 5$, so it is smooth immediately and never occupies the transport channel.

5 Four equivalent coordinate systems

Why this section is here. Different calculations become simple in different coordinates. The point is not to multiply notation, but to move between product, factor-gap, norm, and null coordinates without losing information.

Let

$$w = x + iy = cq + i \frac{q^2 - c^2}{2}, \quad \rho := |w| = \sqrt{x^2 + y^2}. \quad (50)$$

Then

$$\rho = \frac{c^2 + q^2}{2}. \quad (51)$$

The most useful coordinate charts are collected in Table 1.

The central identities are

$$\boxed{U := \rho - y = c^2, \quad V := \rho + y = q^2.} \quad (52)$$

Consequently,

$$x = \sqrt{UV} = cq, \quad y = \frac{V - U}{2}. \quad (53)$$

These null coordinates linearize the arithmetic cutoffs. They also make the meaning of $2ab$ exact:

$$\boxed{Y = 2ab = \frac{V - U}{2},} \quad (54)$$

so the imaginary coordinate is half the separation between the squared canonical factors.

Table 1: Equivalent coordinates for a canonical squared factor point.

Coordinate system	Definition	Exact inverse information
Factor coordinates	(c, q)	$m = cq, q = P^+(m)$
Fermat coordinates	$a = (c + q)/2, b = (q - c)/2$	$c = a - b, q = a + b$
Squared Cartesian	$x = cq, y = (q^2 - c^2)/2$	$w = x + iy = (a + ib)^2$
Radial-null	$U = \rho - y, V = \rho + y$	$U = c^2, V = q^2, x^2 = UV$
Square-root	$a^2 = (\rho + x)/2, b^2 = (\rho - x)/2$	sign of b is sign of y

6 Fixed-cofactor parabolas

Why this section is here. Once the cofactor is fixed, the canonical points are no longer an unstructured cloud. They lie on explicit curves, which lets arithmetic transitions be read geometrically.

Fix a positive cofactor c and let q vary. In squared Cartesian coordinates,

$$x = cq, \quad y = \frac{q^2 - c^2}{2}. \quad (55)$$

Eliminating $q = x/c$ gives the cofactor parabola

$$\boxed{\gamma_c(x) = \frac{x^2 - c^4}{2c^2}}. \quad (56)$$

The canonical point cloud is not the filled region between envelopes. It is the arithmetic sampling

$$x = cq, \quad q \text{ prime}, \quad q > P^+(c), \quad (57)$$

on the family [Equation \(56\)](#).

The curve crosses the real axis at

$$x = c^2. \quad (58)$$

Points to the right have $q > c$ and positive ordinate; points to the left have $q < c$ and negative ordinate.

The $c = 1$ curve is

$$\gamma_1(x) = \frac{x^2 - 1}{2}. \quad (59)$$

Its arithmetic samples are exactly the prime points, since $m = 1 \cdot q = q$. It is the outer positive envelope of the complete canonical cloud. The curve

$$\gamma_3(x) = \frac{x^2 - 81}{18} \quad (60)$$

is the first nontrivial odd-cofactor envelope used in semiprime search.

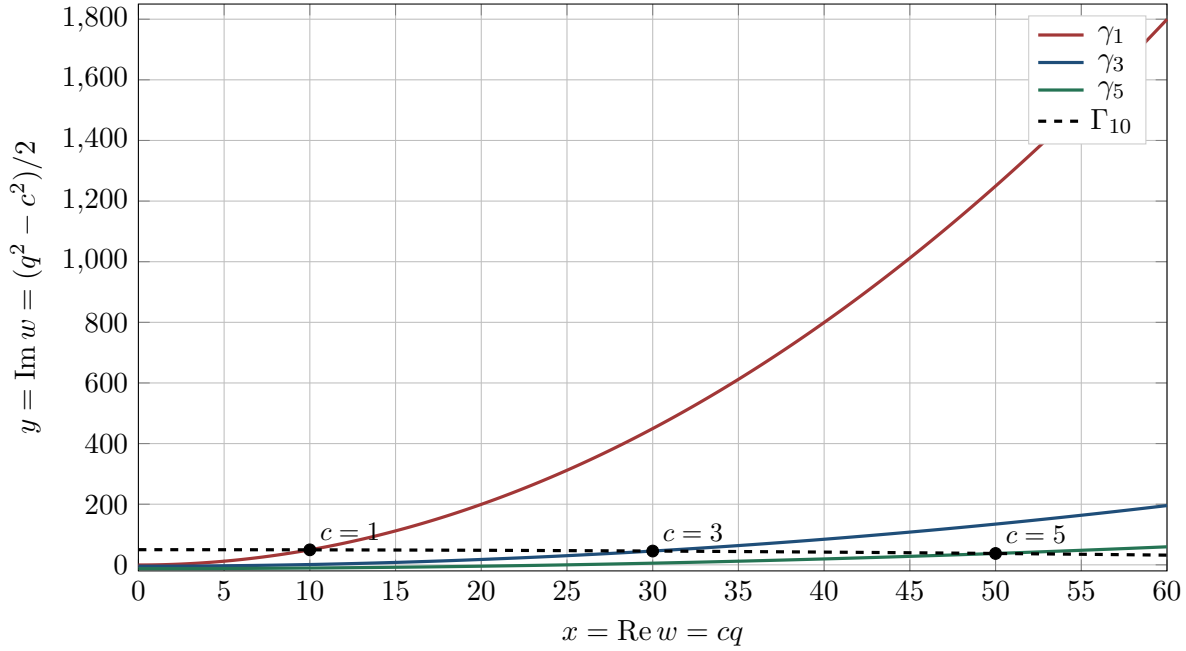


Figure 2: Cofactor parabolas γ_c and the moving cutoff parabola Γ_R for $R = 10$. Each γ_c meets Γ_R at $x = cR$. Actual canonical points are discrete prime samples on the curves.

7 Two moving boundaries

Why this section is here. The decomposition changes with the cutoff. The goal here is to represent that changing arithmetic classification as motion of two simple boundaries through one fixed point cloud.

Fix $R > 0$. The complete arithmetic prefix is

$$\mathcal{P}_R := \{w = x + iy \in \mathcal{W} : x < R^2\}. \quad (61)$$

Since $V = q^2 = |w| + y$, the smooth condition $q \leq R$ is

$$|w| + y \leq R^2. \quad (62)$$

Define

$$\mathcal{A}_R := \{w \in \mathcal{P}_R : |w| + \text{Im } w \leq R^2\}, \quad (63)$$

$$\mathcal{U}_R := \{w \in \mathcal{P}_R : |w| + \text{Im } w > R^2\}. \quad (64)$$

The prefix is the disjoint union

$$\mathcal{P}_R = \mathcal{A}_R \dot{\cup} \mathcal{U}_R. \quad (65)$$

Solving Equation (62) on the boundary gives

$$\Gamma_R(x) = \frac{R^4 - x^2}{2R^2}. \quad (66)$$

Inside the prefix strip $x < R^2$, the smooth region lies below the graph and the transport region lies above it.

In null coordinates, the same regions are even simpler:

$$\mathcal{P}_R: \quad UV < R^4, \quad (67)$$

$$\mathcal{A}_R: \quad UV < R^4, \quad V \leq R^2, \quad (68)$$

$$\mathcal{U}_R: \quad UV < R^4, \quad V > R^2. \quad (69)$$

For a transport point, $UV < R^4 < V^2$, so automatically $U < R^2$. Thus

$$\mathcal{U}_R: \quad U < R^2 < V, \quad UV < R^4. \quad (70)$$

This is exactly $c < R < q$.

7.1 Intersection with every cofactor curve

Proposition 7.1 (Intersection factorization). *For $c, R > 0$,*

$$\gamma_c(x) - \Gamma_R(x) = \frac{(R^2 + c^2)(x^2 - c^2 R^2)}{2c^2 R^2}. \quad (71)$$

Consequently,

$$\gamma_c(x) = \Gamma_R(x) \iff x = cR. \quad (72)$$

At the arithmetic point $x = cq$,

$$\gamma_c(cq) - \Gamma_R(cq) = \frac{(R^2 + c^2)(q^2 - R^2)}{2R^2}. \quad (73)$$

The sign is therefore exact:

$$q < R \iff w_{c,q} \text{ lies below } \Gamma_R, \quad (74)$$

$$q > R \iff w_{c,q} \text{ lies above } \Gamma_R. \quad (75)$$

Proposition 7.2 (Orthogonal crossing). *At the intersection $x = cR$,*

$$\gamma'_c(cR) = \frac{R}{c}, \quad \Gamma'_R(cR) = -\frac{c}{R}, \quad (76)$$

so

$$\gamma'_c(cR)\Gamma'_R(cR) = -1. \quad (77)$$

Hence each cofactor curve meets the moving cutoff orthogonally.

The transition is therefore transverse, never tangent. A canonical point cannot remain trapped in a thin boundary layer as the cutoff passes its largest prime.

8 The moving-region interpretation of $A - T$

Why this section is here. This section translates the algebraic identity $A - T$ into a geometric partition. It is the bridge between the original Mertens sum and the later kernel formulation.

Using the signed measure Equation (49), define

$$\mathcal{A}(R) := \nu(\mathcal{A}_R), \quad \mathcal{U}(R) := \nu(\mathcal{U}_R). \quad (78)$$

At $R = R_n$,

$$\mathcal{A}(R_n) = A_n, \quad \mathcal{U}(R_n) = -T_n. \quad (79)$$

Indeed, the transport points satisfy $c < R_n < q$ and carry signed point mass

$$\mu(cq) = -\mu(c).$$

Therefore

$$\boxed{S_n = \nu(\mathcal{P}_{R_n}) = \mathcal{A}(R_n) + \mathcal{U}(R_n) = A_n - T_n.} \quad (80)$$

Equivalently, define projectors

$$P_R(w) := \mathbf{1}_{\{\operatorname{Re} w < R^2\}}, \quad (81)$$

$$Q_R(w) := \mathbf{1}_{\{|w| + \operatorname{Im} w \leq R^2\}}. \quad (82)$$

Then

$$\mathcal{A}(R) = \int P_R Q_R d\nu, \quad (83)$$

$$\mathcal{U}(R) = \int P_R (1 - Q_R) d\nu, \quad (84)$$

$$S(R) = \int P_R d\nu. \quad (85)$$

The identity is the pointwise partition

$$P_R = P_R Q_R + P_R (1 - Q_R). \quad (86)$$

9 Entry, transition, and telescoping

Why this section is here. A source contributes at different times for different reasons. Separating entry from transition shows which changes are genuine new data and which are internal transfers that cancel.

For a canonical source $m = cq$, define its first square-prefix index

$$e(m) := \lfloor \sqrt{m} \rfloor. \quad (87)$$

This is the least n such that $m < R_n^2$. Define the smoothness-transition index

$$h(m) := q - 1. \quad (88)$$

This is the least n such that $q \leq R_n$.

The transport lifetime is

$$L(m) := (h(m) - e(m))_+. \quad (89)$$

There are two cases:

1. If $q > \sqrt{m}$, equivalently $q > c$ and $Y > 0$, then the point enters above the moving parabola and occupies transport for the discrete interval

$$e(m) \leq n < h(m).$$

2. If $q \leq \sqrt{m}$, equivalently $q \leq c$ and $Y \leq 0$, then $h(m) \leq e(m)$ and the point enters already smooth. Its transport interval is empty.

Define the pointwise indicators

$$p_n(m) := \mathbf{1}_{\{e(m) \leq n\}}, \quad (90)$$

$$u_n(m) := \mathbf{1}_{\{e(m) \leq n < h(m)\}}, \quad (91)$$

$$a_n(m) := \mathbf{1}_{\{e(m) \leq n, h(m) \leq n\}}. \quad (92)$$

Then

$$\boxed{p_n(m) = u_n(m) + a_n(m)}. \quad (93)$$

9.1 Why the crossing cancels

Let E_n^A be the signed mass entering the new vertical strip already below the moving parabola, let E_n^U be the signed mass entering above it, and let C_n be the signed mass crossed by the moving parabola between R_{n-1} and R_n . Then

$$\mathcal{A}(R_n) - \mathcal{A}(R_{n-1}) = E_n^A + C_n, \quad (94)$$

$$\mathcal{U}(R_n) - \mathcal{U}(R_{n-1}) = E_n^U - C_n. \quad (95)$$

Therefore

$$\boxed{S_n - S_{n-1} = E_n^A + E_n^U = \sum_{m \in B_n} \mu(m)}. \quad (96)$$

The entire crossing mass cancels.

This is the telescopic nature of the squared coordinate: the point's imaginary height determines when the moving parabola reaches it, while the crossing itself contributes zero to the increment of the total prefix mass.

10 Why $2ab$ measures both height and lifetime

Why this section is here. The same imaginary coordinate that measures factor imbalance also measures how long a source remains in transport. This is the key space-time link in the construction.

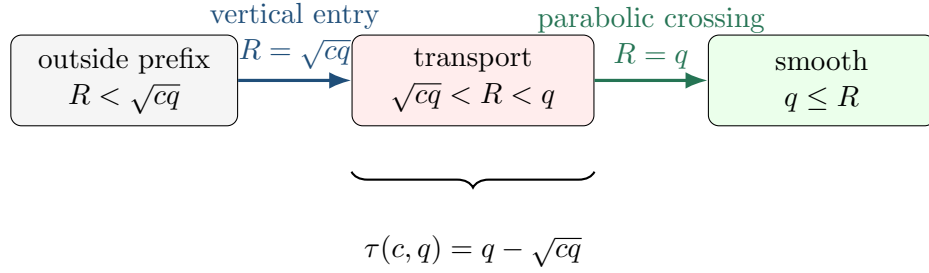


Figure 3: The exact state path of a positive-height canonical point. The second transition only reallocates signed mass between $-T$ and A , so it cancels from the residual increment.

Define the signed continuous delay

$$\tau(c, q) := q - \sqrt{cq}. \quad (97)$$

It is positive exactly for transport-capable points, zero at $c = q$, and negative for points that enter already smooth.

Put

$$r := \frac{c}{q}. \quad (98)$$

Then

$$Y = \frac{q^2}{2}(1 - r^2) \quad (99)$$

and

$$\tau = q(1 - \sqrt{r}). \quad (100)$$

Since

$$1 - r^2 = (1 - \sqrt{r})(1 + \sqrt{r})(1 + r), \quad (101)$$

we obtain an exact identity.

Theorem 10.1 (Exact height–lifetime identity). *For every positive pair (c, q) ,*

$$Y = \frac{q\tau}{2} \left(1 + \sqrt{\frac{c}{q}}\right) \left(1 + \frac{c}{q}\right). \quad (102)$$

Equivalently,

$$\tau = \frac{2Y}{q \left(1 + \sqrt{c/q}\right) (1 + c/q)}. \quad (103)$$

The sign of Y is therefore exactly the sign of the continuous delay. On the transport side $0 < c/q < 1$, the denominator in Equation (103) lies between q and $4q$, giving

$$\frac{Y}{2q} \leq \tau \leq \frac{2Y}{q}. \quad (104)$$

Since

$$L(m) = q - 1 - \lfloor \sqrt{m} \rfloor \leq q - \sqrt{m} = \tau(c, q) \quad (105)$$

for transport points,

$$\boxed{L(m) \leq \frac{2Y_m}{q_m}.} \quad (106)$$

In squared-space coordinates, $q = \sqrt{|w| + \operatorname{Im} w}$, so

$$L(w) \leq \frac{2(\operatorname{Im} w)_+}{\sqrt{|w| + \operatorname{Im} w}}. \quad (107)$$

A point close to the real axis cannot remain in transport for many prefix steps.

11 Low-height clustering in square blocks

Why this section is here. Low height is the part of the problem that geometry alone can control. Here bounded factor gap becomes bounded occupancy, which yields a genuine theorem before any high-sector cancellation is assumed.

The n th square block becomes the vertical strip

$$\mathcal{B}_n := \{w_m : n^2 \leq \operatorname{Re} w_m < (n+1)^2\}. \quad (108)$$

For a canonical pair (c, q) , define the absolute factor gap

$$d := |q - c|. \quad (109)$$

Then

$$|Y| = \frac{|q^2 - c^2|}{2} = \frac{d(q+c)}{2}. \quad (110)$$

For a squarefree source $m > 1$, one has $d \geq 1$: equality $d = 0$ would force $m = c^2$.

Lemma 11.1 (At most one product per positive absolute gap). *Fix $n \geq 1$ and an integer $d \geq 1$. There is at most one unordered positive factor pair $\{c, q\}$ with $|q - c| = d$ and*

$$n^2 \leq cq < (n+1)^2. \quad (111)$$

Proof. Write the smaller factor as r and the larger as $r + d$. The product is $f_d(r) = r(r + d)$. If $f_d(r) \in B_n$, then

$$2r + d \geq 2\sqrt{r(r + d)} \geq 2n.$$

Hence

$$f_d(r+1) - f_d(r) = 2r + d + 1 \geq 2n + 1,$$

which exceeds the integer diameter $2n$ of the block. Thus two distinct values of r cannot both occur. $\square$

Theorem 11.2 (Sharpened absolute low-height clustering). *For every $H \geq 0$,*

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq H\} \leq \left\lfloor \frac{H}{n} \right\rfloor.} \quad (112)$$

Moreover,

$$\boxed{\#\{w_m \in \mathcal{B}_n : |Y_m| \leq n\} = 0.} \quad (113)$$

Proof. If $cq \in B_n$, then

$$c + q \geq 2\sqrt{cq} \geq 2n.$$

Therefore

$$|Y| = \frac{d(c+q)}{2} \geq dn.$$

The condition $|Y| \leq H$ implies $d \leq \lfloor H/n \rfloor$. Since $d \geq 1$, there are at most $\lfloor H/n \rfloor$ possible gaps, and [Lemma 11.1](#) gives at most one point for each gap.

If $|Y| \leq n$, the inequalities force $d = 1$ and $c + q = 2n$. But $|q - c| = 1$ makes $c + q$ odd, whereas $2n$ is even. Hence equality is impossible and the set is empty. $\square$

11.1 Parity-refined bounds

The factor gap has a prescribed parity:

- if m is odd, then c and q are odd, so d is even;
- if m is even and squarefree, then the canonical factors have opposite parity, so d is odd.

Let

$$D := \left\lfloor \frac{H}{n} \right\rfloor. \tag{114}$$

The permitted positive gaps are therefore

$$2, 4, \dots, 2 \left\lfloor \frac{D}{2} \right\rfloor$$

for odd sources and

$$1, 3, \dots, 2 \left\lceil \frac{D}{2} \right\rceil - 1$$

for even sources. Consequently,

$$\#\{w_m \in \mathcal{B}_n : m \text{ odd}, |Y_m| \leq H\} \leq \left\lfloor \frac{H}{2n} \right\rfloor, \tag{115}$$

$$\#\{w_m \in \mathcal{B}_n : m \text{ even}, |Y_m| \leq H\} \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil. \tag{116}$$

At the endpoint $H = n$, both populations are empty by [Equation \(113\)](#).

At height $H = \Lambda n$, the scale-free bounds become

$$\#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\} \leq \lfloor \Lambda \rfloor, \tag{117}$$

$$\#\{m \text{ odd} : |Y_m| \leq \Lambda n\} \leq \left\lfloor \frac{\Lambda}{2} \right\rfloor. \tag{118}$$

11.2 A sharper low-height space–time budget

For negative-height points, the transport lifetime is zero. For positive-height points in $\mathcal{B}_n$, one has $q \geq \sqrt{m} \geq n$, so Equation (106) gives

$$L(m) \leq \frac{2H}{n} \quad (0 < Y_m \leq H). \quad (119)$$

Therefore

$$\boxed{\sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{n} \right\rfloor \left(1 + \frac{2H}{n}\right).} \quad (120)$$

The parity-refined versions are

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ odd}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lfloor \frac{H}{2n} \right\rfloor \left(1 + \frac{2H}{n}\right), \quad (121)$$

$$\sum_{\substack{w_m \in \mathcal{B}_n \\ m \text{ even}, |Y_m| \leq H}} (1 + L(m)) \leq \left\lceil \frac{1}{2} \left\lfloor \frac{H}{n} \right\rfloor \right\rceil \left(1 + \frac{2H}{n}\right). \quad (122)$$

At $H = \Lambda n$, all three right sides are bounded independently of n .

12 Conformal expansion and the missing two powers

Why this section is here. The square map explains the scale but not the sign. Its Jacobian supplies the required two powers of dilution; the later Gram analysis must still explain cancellation.

Consider the continuous map

$$\Phi(c, q) := (x(c, q), y(c, q)) = \left(cq, \frac{q^2 - c^2}{2}\right). \quad (123)$$

Its differential is

$$D\Phi(c, q) = \begin{pmatrix} q & c \\ -c & q \end{pmatrix}. \quad (124)$$

The two columns are orthogonal and have equal length $\sqrt{c^2 + q^2}$. Consequently,

$$D\Phi(c, q)^\top D\Phi(c, q) = (c^2 + q^2)I. \quad (125)$$

Thus Φ is conformal: locally it rotates and expands every direction by the same factor

$$\lambda(c, q) = \sqrt{c^2 + q^2}. \quad (126)$$

The area Jacobian is

$$\boxed{J(c, q) = \det D\Phi(c, q) = c^2 + q^2 = 2|w|.} \quad (127)$$

By AM–GM,

$$c^2 + q^2 \geq 2cq = 2x. \quad (128)$$

Therefore, in the square strip $x \asymp n^2$,

$$\lambda(c, q) \gtrsim n, \quad J(c, q) \gtrsim n^2. \quad (129)$$

This is the two-dimensional version of the vertical spacing estimate. Unit-scale separation in the factor plane becomes scale- n separation in the squared plane, while unit factor-lattice area becomes scale- n^2 area. The coherent square-prefix ceiling loses exactly two powers if the relevant kernel neighborhoods have bounded squared-space area after localization.

Caution 12.1 (What the Jacobian does not prove). The Möbius point cloud is discrete and arithmetically restricted, and the square-prefix kernel is not Euclidean convolution. The Jacobian therefore does not by itself prove a signed variance estimate. It proves the exact geometric dilution scale that a valid operator argument must inherit.

12.1 Two-dimensional area versus one-dimensional cofactor curves

The conformal Jacobian governs two-dimensional regions in the continuous (c, q) factor plane. The arithmetic measure is not two-dimensional: for each fixed cofactor c , it is supported on the one-dimensional prime-sampled curve

$$q \mapsto \Phi(c, q) = \left(cq, \frac{q^2 - c^2}{2} \right), \quad q \text{ prime}, \quad q > P^+(c).$$

Along that curve,

$$\left\| \frac{\partial \Phi}{\partial q}(c, q) \right\| = \sqrt{c^2 + q^2}. \quad (130)$$

Thus one-dimensional arclength is also expanded, but the number and arithmetic placement of samples are governed by primes rather than by the area Jacobian. A two-dimensional density estimate cannot be applied directly to a single cofactor curve.

The outer curve $c = 1$ makes this obstruction explicit. Its contribution to the n th complete prefix is

$$S_n^{(1)} = -\pi(X_n), \quad (131)$$

because every prime $q \leq X_n$ has canonical pair $(1, q)$ and Möbius weight -1 . The prime number theorem therefore predicts

$$\sum_{n \leq N} |S_n^{(1)}|^2 \sim \frac{N^5}{20 \log^2 N}, \quad (132)$$

up to the usual slowly varying correction. The prime curve by itself is not on the cubic scale. Cancellation between $c = 1$ and the aggregate of the curves $c \geq 2$ is indispensable.

This also shows why a bound of the form

$$\sum_{q \leq Q} e\left(\alpha \frac{q^2 - c^2}{2}\right) \ll Q^{1/2+\varepsilon}$$

cannot hold uniformly in α : at $\alpha = 0$ the left side is $\pi(Q) + O(1)$. The zero and major-arc modes must be cancelled across cofactor curves by the Möbius weights $\mu(c)$ and the exact smooth-transport architecture. Oscillatory estimates are relevant only after that coherent component has been isolated.

12.2 Small-modulus prime resonances

The coherent rational modes are more rigid than a generic major-arc heuristic suggests.

Theorem 12.2 (Prime squares modulo 24). *If $q > 3$ is prime, then*

$$\boxed{q^2 \equiv 1 \pmod{24}}. \quad (133)$$

Proof. The prime q is odd, so $q^2 \equiv 1 \pmod{8}$. Also $3 \nmid q$, so $q \equiv \pm 1 \pmod{3}$ and $q^2 \equiv 1 \pmod{3}$. Since 8 and 3 are coprime, the Chinese remainder theorem gives Equation (133). $\square$

Corollary 12.3 (Perfect coherence on the 2–3 resonance lattice). *Let $m \in \mathbb{Z}$ and put $\alpha = m/12$. For every prime $q > 3$,*

$$e\left(\alpha \frac{q^2}{2}\right) = e\left(\frac{mq^2}{24}\right) = e\left(\frac{m}{24}\right). \quad (134)$$

Therefore, if $Q > 3$,

$$\boxed{\sum_{\substack{Q < q \leq 2Q \\ q \text{ prime}}} e\left(\alpha \frac{q^2}{2}\right) = e\left(\frac{m}{24}\right) (\pi(2Q) - \pi(Q))}. \quad (135)$$

In particular, the magnitude equals the full prime count.

This exact coherence includes

$$\alpha \in \left\{0, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}, \frac{1}{6}, \frac{5}{6}\right\}.$$

It is not a failure of the major/minor-arc philosophy. It identifies a finite, explicit resonant component that must be reproduced by the major-arc model and then cancelled across cofactor curves.

Proposition 12.4 (The rational quadratic phase has modulus $2r$). *Let $a, r, u \in \mathbb{Z}$ with $r \neq 0$. Then*

$$\frac{e(a(u+r)^2/(2r))}{e(au^2/(2r))} = (-1)^{ar}. \quad (136)$$

Consequently $u \mapsto e(au^2/(2r))$ is always periodic modulo $2r$, but when ar is odd it is not a well-defined function of $u \bmod r$.

Proof. Expanding the square gives

$$\frac{a(u+r)^2 - au^2}{2r} = au + \frac{ar}{2}.$$

The integer term au disappears under $e(t) = e^{2\pi it}$, while $e(ar/2) = (-1)^{ar}$. $\square$

Definition 12.5 (Corrected reduced quadratic prime factor). For $r \geq 1$ and $(a, r) = 1$, define

$$G_{2r}^\times(a) := \sum_{\substack{u \bmod 2r \\ (u, 2r)=1}} e\left(\frac{au^2}{2r}\right), \quad (137)$$

$$\gamma(a, r) := \frac{G_{2r}^\times(a)}{\varphi(2r)}. \quad (138)$$

At $a = 1, r = 3$, the units modulo 6 are 1 and 5, both with square 1 modulo 6. Thus

$$G_6^\times(1) = 2e(1/6), \quad \boxed{\gamma(1, 3) = e(1/6), \quad |\gamma(1, 3)| = 1.} \quad (139)$$

By contrast, the expression obtained by summing representatives modulo 3 is

$$e(1/6) + e(4/6) = 0.$$

That vanishing is an artifact of using a modulus on which the phase is not well-defined.

Caution 12.6 (Cell-mask Fourier mass and prime-phase coherence are different). The ideal prime-3 cell mask $(+1, -1, +1)$ has mean amplitude $1/3$ and mean-mode energy $1/9$. This is a statement in the cell-index Fourier variable. It is not a bound for the quadratic prime exponential sum at $\alpha = 1/3$, whose corrected rational factor has magnitude 1 by [Equation \(139\)](#). The coherent prime-phase mode must first be represented explicitly and cancelled across the Möbius-weighted cofactor channels; only the complementary component is eligible for the cell-mask contraction estimate.

Definition 12.7 (Dyadic squared-cofactor exponential sum). For dyadic $C, Q \geq 1$, define

$$F_{C,Q}(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \sum_{\substack{Q < q \leq 2Q \\ q \text{ prime} \\ q > P^+(c)}} e\left(\alpha \frac{q^2}{2}\right). \quad (140)$$

A concrete analytic route is a major/minor-arc decomposition for $F_{C,Q}$, coupled to the Fourier representation of localized squared-space kernels. The major arcs must reproduce and cancel the coherent cofactor-curve main terms, while the minor arcs must satisfy a large-sieve or mean-square estimate. This is the analytic content that the conformal geometry identifies but does not itself prove.

13 Vertical spacing over a fixed product

Why this section is here. The fixed-product fibre gives a direct spacing law. These estimates prevent low-height points from accumulating too densely above one square block.

Fix $m > 0$ and parameterize the full factor hyperbola by signed $b \in \mathbb{R}$:

$$a = \sqrt{m + b^2}, \quad Y_m(b) = 2b\sqrt{m + b^2}. \quad (141)$$

Then

$$Y'_m(b) = 2 \frac{m + 2b^2}{\sqrt{m + b^2}} \geq 2\sqrt{m}. \quad (142)$$

The derivative is positive for all signed b , so the map is strictly increasing from the lower to the upper half-plane.

If admissible b values differ by at least h , then

$$|\Delta Y| \geq 2h\sqrt{m}. \quad (143)$$

At square scale $m \asymp n^2$, vertical spacing is at least order n . This agrees with the conformal scale factor in [Equation \(129\)](#).

14 Square-prefix energy as a signed kernel

Why this section is here. Energy bounds are quadratic statements. Rewriting the prefix variance as a kernel makes the diagonal and off-diagonal contributions visible and prepares the signed Gram formulation.

For a canonical point w_m , define

$$p_n(w_m) := \mathbf{1}_{\{e(m) \leq n\}}. \quad (144)$$

Then

$$S_n = \int p_n(w) d\nu(w). \quad (145)$$

For a terminal index N , define

$$K_N(w, w') := \sum_{n=1}^N p_n(w) p_n(w'). \quad (146)$$

Since each p_n is a tail indicator,

$$K_N(w, w') = (N - \max\{e(w), e(w')\} + 1)_+. \quad (147)$$

Therefore

$$\sum_{n=1}^N |S_n|^2 = \iint_{\mathcal{W} \times \mathcal{W}} K_N(w, w') d\nu(w) d\nu(w'). \quad (148)$$

The channel indicators satisfy

$$p_n = a_n + u_n, \quad (149)$$

so the kernel splits exactly into smooth–smooth, transport–transport, and two cross terms.

For transport, the overlap kernel has the interval form

$$K_N^{UU}(w, w') = \#([e(w), h(w)) \cap [e(w'), h(w')] \cap [1, N]). \quad (150)$$

Consequently,

$$K_N^{UU}(w, w') \leq \min\{L(w), L(w')\}. \quad (151)$$

The imaginary coordinate therefore controls temporal support through [Equation \(107\)](#).

15 The proved cubic bound at low height

Why this section is here. This is the closed part of the analytic argument: the low-height sector already obeys the cubic scale for geometric reasons.

Fix $\Lambda > 0$. Assign each source m to its entry block

$$j = e(m) = \lfloor \sqrt{m} \rfloor \quad (152)$$

and define the low-height block increment

$$d_j^{\text{low}} := \sum_{\substack{m \in B_j \\ |Y_m| \leq \Lambda j}} \mu(m). \quad (153)$$

By Equation (117),

$$|d_j^{\text{low}}| \leq \lfloor \Lambda \rfloor. \quad (154)$$

Define

$$S_n^{\text{low}} := \sum_{j \leq n} d_j^{\text{low}}. \quad (155)$$

Even under complete sign coherence,

$$|S_n^{\text{low}}| \leq \lfloor \Lambda \rfloor n. \quad (156)$$

Hence

$$\boxed{\sum_{n \leq N} |S_n^{\text{low}}|^2 \ll_{\Lambda} N^3.} \quad (157)$$

No Möbius cancellation is used.

Let

$$S_n = S_n^{\text{low}} + S_n^{\text{high}}. \quad (158)$$

Then

$$\sum_{n \leq N} |S_n|^2 \leq 2 \sum_{n \leq N} |S_n^{\text{low}}|^2 + 2 \sum_{n \leq N} |S_n^{\text{high}}|^2. \quad (159)$$

More importantly, on every translated window $I = [N, N + H)$ with $1 \leq H \leq N$,

$$\boxed{\sum_{n=N}^{N+H-1} |S_n^{\text{low}}|^2 \leq 4 \lfloor \Lambda \rfloor^2 H N^2.} \quad (160)$$

Indeed, $n < 2N$ throughout the window and $|S_n^{\text{low}}| \leq \lfloor \Lambda \rfloor n$. Since $S^{\text{high}} = S - S^{\text{low}}$, the two norm inequalities give

$$V_{\text{tot}}(N, H) \leq 2V_{\text{low}}(N, H) + 2V_{\text{high}}(N, H), \quad (161)$$

$$V_{\text{high}}(N, H) \leq 2V_{\text{tot}}(N, H) + 2V_{\text{low}}(N, H). \quad (162)$$

Thus the low-height sector is harmless at the exact local RH scale, and the total and high-sector growth criteria are equivalent without estimating their energy cross term. The entire unresolved problem is therefore localized, equivalently, to the high-height point cloud.

16 The unresolved high-height problem

17 The unresolved high-height problem

Why this section is here. Everything difficult is concentrated here. The high sector cannot be controlled shell by shell; it requires cross-curve and cross-shell cancellation inside one operator estimate.

The high sector consists of canonical points satisfying

$$|Y_m| > \Lambda e(m). \quad (163)$$

These points lie far from the real axis relative to their square scale, but they remain highly structured:

1. they lie on the explicit curves γ_c ;
2. their horizontal coordinates are prime samples $x = cq$ with $q > P^+(c)$;
3. their one-dimensional curve spacing is expanded by $\sqrt{c^2 + q^2}$;
4. two-dimensional factor regions are expanded by $c^2 + q^2$;
5. positive-height points have exact transport intervals $[e(m), h(m))$;
6. each curve crosses every moving cutoff exactly once and orthogonally.

For each squarefree cofactor c , define the signed curve measure

$$\nu_c := -\mu(c) \sum_{\substack{q \text{ prime} \\ q > P^+(c)}} \delta_{cq+i(q^2-c^2)/2}. \quad (164)$$

Then

$$\nu = \delta_1 + \sum_{c \geq 1} \nu_c, \quad (165)$$

and the variance decomposes as

$$\sum_{c, c'} \iint K_N(w, w') d\nu_c(w) d\nu_{c'}(w'). \quad (166)$$

The curve $c = 1$ is not a negligible boundary case. By [Equations \(131\) and \(132\)](#), its unsigned energy is of order $N^5 / \log^2 N$. The cubic residual therefore requires near-complete cancellation between this prime curve and the remaining cofactor curves. The numerical results in [Section 20](#) measure that cancellation directly.

17.1 The kernel-weighted large-sieve target

Fix a local prefix interval

$$I = [N, N + H), \quad 1 \leq H \leq N, \quad (167)$$

and dyadic ranges $C < c \leq 2C$, $Q < q \leq 2Q$. Let $\omega_{I, \ell}(c, q)$ be a smooth localization to the prefix interval, the transport interval, and a high vertical shell ℓ . Define

$$F_{C, Q, \ell}^I(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \sum_{\substack{Q < q \leq 2Q \\ q \text{ prime} \\ q > P^+(c)}} \omega_{I, \ell}(c, q) e\left(\alpha \frac{q^2}{2}\right). \quad (168)$$

Let $\widehat{K}_{I, \ell}(\alpha) \geq 0$ be the Fourier multiplier of the corresponding localized positive semidefinite prefix/transport-overlap kernel.

For parameters $R_0, \Delta \geq 1$, define the major arcs

$$\mathfrak{M}(R_0, \Delta) := \bigcup_{1 \leq r \leq R_0} \bigcup_{\substack{0 \leq a < r \\ (a, r) = 1}} \left\{ \alpha \in \mathbb{R}/\mathbb{Z} : \left| \alpha - \frac{a}{r} \right| \leq \frac{\Delta}{rN^2} \right\}. \quad (169)$$

For $\alpha = a/r + \beta$ on such an arc, the rational quadratic phase is periodic modulo $2r$ by [Proposition 12.4](#). The explicit Hardy–Littlewood model must therefore replace the prime indicator in [Equation \(168\)](#) by its reduced-residue main term modulo $2r$:

$$\mathcal{M}_{C,Q,\ell}^I(\alpha) := \sum_{\substack{C < c \leq 2C \\ \mu(c) \neq 0}} -\mu(c) e\left(-\alpha \frac{c^2}{2}\right) \gamma(a, r) \int_Q^{2Q} \omega_{I,\ell}(c, t) e\left(\beta \frac{t^2}{2}\right) \frac{dt}{\log t}, \quad (170)$$

$$\gamma(a, r) := \frac{1}{\varphi(2r)} \sum_{\substack{u \bmod 2r \\ (u, 2r) = 1}} e\left(\frac{au^2}{2r}\right). \quad (171)$$

Primes dividing $2r$ are treated as explicit exceptional terms; they are absent whenever the prime range begins above $2R_0$. Outside $\mathfrak{M}(R_0, \Delta)$, put $\mathcal{M}_{C,Q,\ell}^I = 0$.

Let $R_0 = R_0(M) = M^{o(1)}$ be the major-arc denominator cutoff at packet scale M . Let $P_{\text{res},M}$ denote the orthogonal projection in the localized packet Hilbert space onto the span of the corrected rational major-arc modes with $r \leq R_0(M)$, and put $P_{\text{nr},M} = I - P_{\text{res},M}$. This is a finite-dimensional projection at each scale, but it is not one fixed finite projection for all scales. For the compressed signed packet state Z_M , write

$$Z_M = Z_M^{\text{res}} + Z_M^{\text{nr}}, \quad Z_M^{\text{res}} = P_{\text{res},M} Z_M, \quad Z_M^{\text{nr}} = P_{\text{nr},M} Z_M. \quad (172)$$

At each fixed scale,

$$\boxed{\|Z_M\|^2 = \|Z_M^{\text{res}}\|^2 + \|Z_M^{\text{nr}}\|^2}. \quad (173)$$

The resonant term is controlled only after summing the corrected major-arc models across the Möbius-weighted cofactor ranges. The nonresonant term is the proper input to minor-arc and cell-mask contraction estimates. Because $P_{\text{res},M}$ changes with M , scale descent can transfer mass between the two components; this leakage must be included in the recursive state rather than silently discarded.

Conjecture 17.1 (Joint kernel-weighted Squared-Cofactor Large Sieve). *For every $\varepsilon > 0$, there are choices $R_0 = N^{o(1)}$ and $\Delta = N^{o(1)}$ such that, uniformly for $1 \leq H \leq N$,*

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{C,Q,\ell} (F_{C,Q,\ell}^I(\alpha) - \mathcal{M}_{C,Q,\ell}^I(\alpha)) \right|^2 \widehat{K}_I(\alpha) d\alpha \ll_{\varepsilon} H N^{2+\varepsilon}, \quad (174)$$

$$\int_{\mathbb{R}/\mathbb{Z}} \left| \sum_{C,Q,\ell} \mathcal{M}_{C,Q,\ell}^I(\alpha) \right|^2 \widehat{K}_I(\alpha) d\alpha \ll_{\varepsilon} H N^{2+\varepsilon}. \quad (175)$$

The dyadic sums range only over cells meeting the high sector and the prefix interval. The sums over cofactor ranges, prime ranges, and height shells remain inside the absolute square. Consequently all cross-curve, cross-shell, and cross-frequency terms are retained.

The first estimate is the joint kernel-weighted nonresonant and major-arc error bound. The second is the required cancellation of the corrected explicit major-arc models across cofactor scales, including the exact 2–3 resonance lattice from [Corollary 12.3](#). Both clauses are necessary: at zero frequency, and at several nonzero rational frequencies, the prime curve has full coherent size. A uniform pointwise square-root estimate is therefore false. Equally, the extended numerical experiment below shows that a positive sum of separate shell energies is far too large. The conjecture therefore places the entire signed recombination in the norm generated by the square-prefix kernel.

For a finite shell partition $\{\mathcal{H}_\ell\}$ of the high sector, write

$$S_n^{\text{high}} = \sum_{\ell} S_n^{[\ell]}, \quad G_{\ell m}(N) := \sum_{n \leq N} S_n^{[\ell]} \overline{S_n^{[m]}}. \quad (176)$$

Then

$$\sum_{n \leq N} |S_n^{\text{high}}|^2 = \mathbf{1}^* G(N) \mathbf{1} = \sum_{\ell} G_{\ell\ell}(N) + 2 \operatorname{Re} \sum_{\ell < m} G_{\ell m}(N). \quad (177)$$

Conjecture 17.2 (Joint high-sector signed Gram estimate). *After removing the low-height sector controlled by [Equation \(157\)](#), the full signed shell Gram form satisfies, for every $\varepsilon > 0$,*

$$\mathbf{1}^* G(N) \mathbf{1} \ll_{\varepsilon, \Lambda} N^{3+\varepsilon}. \quad (178)$$

No separate estimate $\sum_{\ell} G_{\ell\ell}(N) \ll N^{3+\varepsilon}$ is assumed or expected. The two clauses of [conjecture 17.1](#) are a sufficient analytic route because they preserve the off-diagonal terms needed for cross-shell and cross-curve cancellation.

This formulation is stronger and more accurate than rewriting RH in the coordinate Y . It asks for control of one joint Gram matrix indexed simultaneously by cofactor ranges, prime ranges, height shells, and rational phase modes. The precise analytic obstruction is not the size of the shell diagonals; it is the uniform signed alignment of their off-diagonal interactions inside the actual-start packet kernel.

18 Why the cubic scale is natural

Why this section is here. Before asking for a proof, we check that every scale is dimensionally consistent. This section explains why N^3 , rather than the coherent N^5 ceiling, is the natural target.

The block increment

$$d_n = S_n - S_{n-1} \quad (179)$$

satisfies the trivial bound $|d_n| \leq 2n + 1$. Expanding the cumulative energy gives

$$\sum_{n \leq N} |S_n|^2 = \sum_{j, k \leq N} (N - \max\{j, k\} + 1) d_j d_k. \quad (180)$$

The fully coherent ceiling is

$$N^5 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{N^2}_{\text{raw pair amplitude}}. \quad (181)$$

The squared map provides the exact missing scale:

$$J(c, q) = c^2 + q^2 \geq 2cq \asymp n^2. \quad (182)$$

Thus a unit area of factor combinations is diluted by n^{-2} in the squared plane. Equivalently, each coordinate direction is stretched by order n . If the localized kernel respects this conformal dilation, then the effective pair amplitude becomes order one:

$$N^3 = \underbrace{N}_{\text{prefix multiplicity}} \times \underbrace{N^2}_{\text{block pairs}} \times \underbrace{1}_{\text{squared-space interaction}}. \quad (183)$$

The low-height theorem proves this reduction directly in the region $|Y| \lesssim n$. The high-height conjecture asks for the corresponding operator statement in the remaining region.

19 The RH-equivalent local criterion

Why this section is here. A global average is not enough to recover pointwise Mertens control. The uniform translated-window estimate is the correct RH-equivalent statement because it allows the decisive choice $H = 1$.

The global cubic energy is the natural benchmark, but a fully pointwise converse is obtained from a uniform local estimate. Define

$$V_{\text{loc}}(N, H) := \sum_{n=N}^{N+H-1} |S_n|^2. \quad (184)$$

Theorem 19.1 (Uniform local square-prefix criterion). *The Riemann Hypothesis is equivalent to the statement that, for every $\varepsilon > 0$,*

$$V_{\text{loc}}(N, H) \ll_{\varepsilon} H N^{2+\varepsilon} \quad (1 \leq H \leq N). \quad (185)$$

Proof. Under RH, the classical Mertens criterion gives

$$M(x) \ll_{\delta} x^{1/2+\delta}$$

for every $\delta > 0$. Since $X_n \asymp n^2$, one has

$$|S_n|^2 \ll_{\varepsilon} n^{2+\varepsilon},$$

and summing H consecutive terms gives Equation (185).

Conversely, set $H = 1$. Then

$$|S_N|^2 \ll_{\varepsilon} N^{2+\varepsilon},$$

so $|S_N| \ll_{\eta} N^{1+\eta}$ for every $\eta > 0$. If

$$n^2 \leq x < (n+1)^2,$$

then

$$|M(x) - M(n^2 - 1)| \leq 2n + 1$$

because $|\mu(m)| \leq 1$. Therefore

$$M(x) \ll_{\eta} x^{1/2+\eta},$$

which is equivalent to RH. □

For the exact low/high split, define

$$V_{\text{low}}(N, H) := \sum_{n=N}^{N+H-1} |S_n^{\text{low}}|^2, \quad (186)$$

$$V_{\text{high}}(N, H) := \sum_{n=N}^{N+H-1} |S_n^{\text{high}}|^2. \quad (187)$$

Theorem 19.2 (Geometric reduction of the RH criterion). *For every fixed $\Lambda > 0$ and every $\varepsilon > 0$,*

$$\boxed{V_{\text{loc}}(N, H) \ll_{\varepsilon} HN^{2+\varepsilon} \iff V_{\text{high}}(N, H) \ll_{\varepsilon, \Lambda} HN^{2+\varepsilon}} \quad (1 \leq H \leq N). \quad (188)$$

Consequently,

$$\boxed{\text{RH} \iff V_{\text{high}}(N, H) \ll_{\varepsilon, \Lambda} HN^{2+\varepsilon} \quad (1 \leq H \leq N)} \quad (189)$$

for every fixed cutoff Λ .

Proof. The local low-sector estimate Equation (160) is $O_{\Lambda}(HN^2)$ and hence is admissible at every $HN^{2+\varepsilon}$ scale. The forward and reverse implications in Equation (188) are exactly Equations (161) and (162). Composing with Theorem 19.1 proves Equation (189). $\square$

Thus the local high-sector signed-Gram estimate is not merely one sufficient route to RH. It is an equivalent geometric formulation. The global cubic estimate in conjecture 17.2 remains a useful benchmark, but the uniform translated-window version of that estimate is the exact RH-equivalent target.

20 Numerical evidence and preregistered diagnostics

Why this section is here. The computations are designed to falsify mechanisms, not to certify RH. Exact checks validate identities; asymptotic diagnostics identify where a proposed uniform theorem would first fail.

The computations reported here use exact integer arithmetic through

$$N_{\text{max}} = 4096, \quad X_{\text{max}} = (4097)^2 - 1 = 16,785,408. \quad (190)$$

A linear sieve produced

$$1,078,362 \text{ primes} \quad \text{and} \quad 10,204,280 \text{ squarefree sources } m \geq 2. \quad (191)$$

20.1 Deterministic exact checks

The implementation independently constructed S_n , A_n , and T_n from the source entry and transition intervals. Over all 4096 prefixes,

$$\max_{n \leq 4096} |S_n - (A_n - T_n)| = 0. \quad (192)$$

Across all 10,204,280 squarefree sources, the following violation counts were also zero:

Table 2: Exact squared-space verification through $X_{\max}$.

Check	Violations
$(q^2 + c^2)^2 = (2m)^2 + (q^2 - c^2)^2$	0
$c^2(q^2 - c^2) = m^2 - c^4$	0
height-lifetime inequality	0
prefix identity $S_n = A_n - T_n$	0

20.2 Small-modulus prime-phase regression test

For the toy interval $2000 < q \leq 4000$, there are exactly

$$\pi(4000) - \pi(2000) = 247 \quad (193)$$

primes. Every one satisfies $q^2 \equiv 1 \pmod{24}$. Hence, for $\alpha = m/12$,

$$S_{\text{prime}}(\alpha) := \sum_{\substack{2000 < q \leq 4000 \\ q \text{ prime}}} e\left(\alpha \frac{q^2}{2}\right) = 247e(m/24), \quad (194)$$

and therefore $|S_{\text{prime}}(\alpha)| = 247$ exactly.

Table 3: Exact prime-phase coherence on the tested 2–3 resonance lattice.

α	common phase	$ S_{\text{prime}}(\alpha) $
0	1	247
1/2	$e(1/4)$	247
1/3	$e(1/6)$	247
2/3	$e(1/3)$	247
1/4	$e(1/8)$	247
3/4	$e(3/8)$	247
1/6	$e(1/12)$	247
5/6	$e(5/12)$	247

The old modulus- r expression fails the regression test at $a = 1, r = 3$:

$$\sum_{\substack{u \bmod 3 \\ (u,3)=1}} e(u^2/6) = e(1/6) + e(4/6) = 0. \quad (195)$$

The corrected units-modulo-6 factor is

$$\frac{1}{\varphi(6)} \sum_{\substack{u \bmod 6 \\ (u,6)=1}} e(u^2/6) = e(1/6), \quad (196)$$

with unit magnitude.

The elevated value at $\alpha = 1/20$ is also arithmetic rather than a numerical artifact. For primes $q > 5$, $q^2 \equiv 1$ or $9 \pmod{40}$. In this interval the class counts are 123 and 124, respectively, so

$$|123e(1/40) + 124e(9/40)| = 199.828062 \dots \quad (197)$$

This illustrates why the complete reduced-square distribution modulo $2r$, not only the denominator r , belongs in the rational major-arc model.

20.3 Smooth–transport matching

Table 4: Exact cumulative prefix energies. Every entry is computed from all prefixes $1 \leq n \leq N$.

N	$\ A\ ^2/N^3$	$\ T\ ^2/N^3$	$\langle A, T \rangle/N^3$	$\ A - T\ ^2/N^3$	S_N
128	0.490490	0.469682	0.475665	0.008842	−18
256	1.386025	1.338898	1.356626	0.011671	4
512	3.709722	3.668905	3.685593	0.007442	38
1024	9.831997	9.807089	9.814616	0.009853	288
2048	26.661079	26.658451	26.655516	0.008497	263
4096	72.821788	73.064243	72.938028	0.009974	329

The residual remains near the cubic scale while the two component energies increase substantially.

20.4 The prime curve does not satisfy the cubic bound

Let

$$S_n^{(1)} = -\pi(X_n)$$

be the contribution of the curve $c = 1$, and put

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}.$$

Then

$$\|S\|^2 = \|S^{(1)}\|^2 + \|S^{(\geq 2)}\|^2 + 2\langle S^{(1)}, S^{(\geq 2)} \rangle. \quad (198)$$

Table 5: Prime-curve versus remaining-curve energy decomposition.

N	V/N^3	$V_{c=1}/N^3$	$V_{c \geq 2}/N^3$	$2\langle c = 1, c \geq 2 \rangle/N^3$	$V/V_{c=1}$	correlation
128	0.008842	51.225450	50.965559	−102.182168	1.7260×10^{-4}	−0.999916714
256	0.011671	146.590116	146.035583	−292.614028	7.9618×10^{-5}	−0.999961911
512	0.007442	441.694929	441.211486	−882.898974	1.6848×10^{-5}	−0.999991721
1024	0.009853	1382.291036	1381.948964	−2764.230146	7.1282×10^{-6}	−0.999996443
2048	0.008497	4450.731250	4450.668331	−8901.391084	1.9091×10^{-6}	−0.999999045
4096	0.009974	14652.537064	14655.994904	−29308.521994	6.8072×10^{-7}	−0.999999667

At $N = 4096$, the prime curve has energy more than $1.46 \times 10^4 N^3$, while the total residual is approximately $9.97 \times 10^{-3} N^3$. The squared norm of the total is only 6.81×10^{-7} of the prime-curve squared norm. This is direct numerical evidence that the central phenomenon is cancellation *between* cofactor curves, not cubic behavior of the prime curve in isolation.

20.5 The dyadic cofactor cascade

For $C \geq 1$, define the partial cofactor residual

$$R_C(x) := 1 - \sum_{1 \leq c \leq C} \mu(c) \Pi_c(x), \quad (199)$$

and its square-prefix energy

$$V_C(N) := \sum_{n \leq N} |R_C(X_n)|^2. \quad (200)$$

Thus

$$R_1(x) = 1 - \pi(x),$$

and $R_C(x) = M(x)$ once C contains every canonical cofactor that can occur below x . The cascade tests how successive dyadic cofactor ranges cancel the prime curve.

Table 6: Complete dyadic cofactor cascade at $N = 4096$. The third column is $V_C(4096)/4096^3$; the fourth is V_C/V_1 . The final row is exact because the largest observed canonical cofactor is $692,835 < 2^{20}$.

C	$R_C(X_{4096})$	V_C/N^3	V_C/V_1
1	-1,078,361	14652.492160	1.000e+00
2	-513,936	3318.367550	2.265e-01
4	-127,056	198.352150	1.354e-02
8	86,580	99.704527	6.805e-03
16	-4,620	0.148513	1.014e-05
32	162,415	344.808298	2.353e-02
64	51,754	36.470907	2.489e-03
128	75,937	76.897452	5.248e-03
256	67,189	59.850472	4.085e-03
512	81,166	84.906888	5.795e-03
1,024	71,179	60.275193	4.114e-03
2,048	43,260	15.704342	1.072e-03
4,096	3,633	4.361902	2.977e-04
8,192	-39,326	14.649760	9.998e-04
16,384	-15,331	1.552681	1.060e-04
32,768	-2,326	0.051346	3.504e-06
65,536	4,622	0.142635	9.735e-06
131,072	708	0.011615	7.927e-07
262,144	206	0.009590	6.545e-07
524,288	332	0.009980	6.811e-07
1,048,576	329	0.009974	6.807e-07

The cascade is strongly nonmonotone. The first four cofactor scales reduce the normalized energy from 14652.49 at $C = 1$ to 0.1485 at $C = 16$, but the next dyadic block raises it to 344.81 at $C = 32$. Subsequent blocks repeatedly destroy and recreate coherent mass before the residual settles near the final cubic scale. This rules out a simple monotone “more cofactors means more cancellation” picture. The analytic estimate must control a signed cascade of dyadic cross-terms.

The machine-readable file `cofactor_cascade.csv` reports every dyadic C for all six prefix checkpoints $N = 128, 256, 512, 1024, 2048, 4096$.

20.6 The dyadic Gram matrix and the $16 < c \leq 32$ spike

Let B_j denote the complete square-prefix vector contributed by the cofactor block

$$2^{j-1} < c \leq 2^j,$$

with $B_0 = P$ for $c = 1$. The exact Gram matrix is

$$G_{ij}(N) := \langle B_i, B_j \rangle_{1 \leq n \leq N}. \quad (201)$$

The following table compares the harmonic Möbius coefficient

$$a_j = \sum_{2^{j-1} < c \leq 2^j} \frac{\mu(c)}{c}$$

from [Theorem 2.5](#) with the finite-sample regression coefficient

$$\hat{\beta}_j := \frac{\langle P, B_j \rangle}{\|P\|^2}.$$

Table 7: Dyadic cofactor block geometry at $N = 4096$. The close agreement of signs, the nearly unit correlations, and the Gram spectrum below exhibit the major-arc rank-one structure.

block	a_j	$\hat{\beta}_j$	$\text{corr}(P, B_j)$	$\ B_j\ ^2/N^3$	$\Delta V/N^3$
$1 < c \leq 2$	-0.500000	-0.524110	-0.999999	4024.9308	-11334.1246
$2 < c \leq 4$	-0.333333	-0.359551	-0.999997	1894.2410	-3120.0154
$4 < c \leq 8$	-0.176190	-0.198795	-0.999993	579.0658	-98.6476
$8 < c \leq 16$	0.070263	0.085079	0.999979	106.0651	-99.5560
$16 < c \leq 32$	-0.123472	-0.155995	-0.999970	356.5833	344.6598
$32 < c \leq 64$	0.078716	0.103523	0.999952	157.0458	-308.3374

For the block $B_{16,32}$ one finds

$$\frac{\|B_{16,32}\|^2}{N^3} = 356.583335, \quad (202)$$

$$\frac{\langle P, B_{16,32} \rangle}{N^3} = -2285.725157, \quad (203)$$

$$\text{corr}(P, B_{16,32}) = -0.999970. \quad (204)$$

Thus the proposed balance $\langle P, B_{16,32} \rangle \approx -\frac{1}{2}\|B_{16,32}\|^2$ is not what occurs. Instead,

$$\boxed{\langle P, B_{16,32} \rangle = -6.410073 \|B_{16,32}\|^2 \text{ at } N = 4096.} \quad (205)$$

The block is almost a negative scalar multiple of the prime curve, as predicted by the rank-one theorem.

The cascade spike has a different cause. Let R_{16} be the cumulative residual after the blocks through $c \leq 16$. Then

$$\frac{\langle R_{16}, B_{16,32} \rangle}{N^3} = -5.961775, \quad (206)$$

so

$$\frac{V_{32} - V_{16}}{N^3} = \frac{\|B_{16,32}\|^2 + 2\langle R_{16}, B_{16,32} \rangle}{N^3} \quad (207)$$

$$= 356.583335 - 11.923550 \quad (208)$$

$$= 344.659785. \quad (209)$$

By $C = 16$, the prime-like coherent mode has already been almost annihilated. The block $16 < c \leq 32$ therefore reintroduces that mode rather than cancelling it. The next block $32 < c \leq 64$ has

$$\frac{\langle R_{32}, B_{32,64} \rangle}{N^3} = -232.691603$$

and lowers the energy by $308.337391N^3$, repairing most of the spike.

The Gram matrix itself is nearly rank one. For the six blocks through $c \leq 32$, its largest eigenvalue contains

$$0.999998101 \quad (210)$$

of the trace and $\lambda_2/\lambda_1 = 1.8926 \times 10^{-6}$. Through $c \leq 64$, the corresponding values are 0.999997519 and 2.4696×10^{-6} . Restricting to the high sector $|Y| > 16n$ changes these figures only to 0.999998118 and 1.8753×10^{-6} through $c \leq 32$. The observed rank-one mode is therefore a high-sector phenomenon, not an artifact of the already controlled low-height points.

20.7 Independent extension through $N = 2^{14}$

A separate memory-limited computation extended the exact checks and structural diagnostics to

$$N = 2^{14} = 16384, \quad X_N = (16385)^2 - 1 \approx 2.68 \times 10^8. \quad (211)$$

The larger targets $N = 2^{16}$ and 2^{17} would require sieving past approximately 4.3×10^9 and 1.7×10^{10} , respectively, and were not attempted in the available memory. No values at those scales are claimed.

The independently constructed smooth, transport, and residual values agree exactly at every tested checkpoint. At the terminal checkpoint,

$$\boxed{A_{16384} = -851896, \quad T_{16384} = -850299, \quad S_{16384} = -1597 = A_{16384} - T_{16384}.} \quad (212)$$

For a cofactor block $C < c \leq 2C$ with $C \asymp N^\sigma$, define the theorem-predicted rank-one residual

$$E_C^{\text{th}} := B_C - \lambda_\sigma a_C P, \quad \lambda_\sigma = \frac{2}{2 - \sigma}. \quad (213)$$

This is distinct from the orthogonal least-squares residual $B_C - \hat{\beta}_C P$. The following table reports the former.

Table 8: Theorem-predicted rank-one residuals at $N = 16384$. Residual energy is four to five orders of magnitude below the raw block energy throughout the tested range.

cofactor block	a_C	$\ B_C\ ^2/N^3$	$\ E_C^{\text{th}}\ ^2/N^3$	residual/raw
(1, 2]	-0.50000	45202.3	0.5166	1.1×10^{-5}
(2, 4]	-0.33333	21074.3	2.4454	1.2×10^{-4}
(4, 8]	-0.17619	6344.8	1.0401	1.6×10^{-4}
(8, 16]	+0.07026	1133.5	0.0481	4.2×10^{-5}
(16, 32]	-0.12347	3752.6	0.1394	3.7×10^{-5}
(32, 64]	+0.07872	1626.2	0.4517	2.8×10^{-4}
(64, 128]	-0.01655	77.4	0.0547	7.1×10^{-4}
(128, 256]	+0.00512	8.9	0.0052	5.8×10^{-4}
(256, 512]	-0.00911	29.4	0.0058	2.0×10^{-4}

The height-shell decomposition is more decisive. With shells defined by the ratio $|Y_m|/j$, where j is the entry block, the diagonal shell energies are:

Using the rounded diagonal values in [Table 9](#),

$$\sum_{\ell} G_{\ell\ell}(N)/N^3 \approx 135.084, \quad 2 \operatorname{Re} \sum_{\ell < m} G_{\ell m}(N)/N^3 \approx -135.074. \quad (214)$$

Thus the high sector is not small by shell. Its observed smallness is produced by nearly complete cancellation of large off-diagonal shell interactions. Any proof that takes absolute values or sums separate shell norms before recombination destroys the principal mechanism visible in the data.

Table 9: Height-shell diagonal energies at $N = 16384$. The recombined energy is not the sum of these rows; it includes their large signed cross terms.

shell for $ Y_m /j$	$\sum_n S_n^{[\ell]} ^2/N^3$
[0, 5)	0.00057
[5, 20)	0.0147
[20, 80)	0.259
[80, 320)	3.81
[320, 1280)	44.7
[1280, ∞)	86.3
signed recombination	0.0104

Exact and secondary reduced-square modes

The exact congruence $q^2 \equiv 1 \pmod{24}$ was verified on all 4459 primes in the tested prime sample. Modulo 40, the same sample occupied exactly the two classes

$$q^2 \equiv 1 \text{ or } 9 \pmod{40}, \quad (215)$$

with counts 2224 and 2235. This is forced for primes $q > 5$ by $q^2 \equiv 1 \pmod{8}$ and $q^2 \equiv 1 \text{ or } 4 \pmod{5}$.

Proposition 20.1 (Two reduced-square classes modulo 40). *For every prime $q > 5$,*

$$\boxed{q^2 \equiv 1 \text{ or } 9 \pmod{40}.} \quad (216)$$

Accordingly, a conductor dividing 40 produces a two-mode prime phase rather than a featureless minor-arc sum. In the tested sample,

frequency	observed prime-sum magnitude
1/20	3607
3/20	1378
1/40	approximately 8–14
9/40	approximately 8–14

The same two exact phase modes can therefore interfere constructively or destructively. Here “resonant” means low-rank exact phase support, not necessarily large amplitude.

Cumulative Möbius-weighted cofactor phase magnitudes through the six blocks ending at $c = 64$ were

θ	(1, 2]	(2, 4]	(4, 8]	(8, 16]	(16, 32]	(32, 64]
1/12	1.0	1.6	3.4	4.4	8.0	13.9
1/20	1.0	1.8	2.5	5.2	7.3	10.7
1/40	1.0	2.0	1.8	2.2	0.38	2.8

(217)

The 1/40 row contains a substantial cancellation event at the block $16 < c \leq 32$, emphasizing again that the resonant main term is a signed cofactor cascade rather than a monotone positive contribution.

20.8 Low-height clustering and lifetime

For fixed Λ , define

$$C_n(\Lambda) := \#\{w_m \in \mathcal{B}_n : |Y_m| \leq \Lambda n\}, \quad (218)$$

$$I_n(\Lambda) := \sum_{\substack{w_m \in \mathcal{B}_n \\ |Y_m| \leq \Lambda n}} (1 + L(m)). \quad (219)$$

The sharpened exact bounds are

$$C_n(\Lambda) \leq \lfloor \Lambda \rfloor, \quad (220)$$

$$I_n(\Lambda) \leq \lfloor \Lambda \rfloor (1 + 2\Lambda), \quad (221)$$

with $C_n(1) = I_n(1) = 0$. The odd and even source counts are bounded by $\lfloor \Lambda/2 \rfloor$ and $\lceil \lfloor \Lambda \rfloor / 2 \rceil$, respectively.

Table 10: Observed low-height maxima over $128 \leq n \leq 4096$, with the sharpened general and parity bounds.

Λ	count	bound	total max	odd bound	odd max	even bound	even max	incidence bound	incidence max
1	0		0	0	0	0	0	0	0
2	2		1	1	0	1	1	10	1
4	4		2	2	1	2	1	36	4
8	8		4	4	2	4	2	136	10
16	16		6	8	3	8	3	528	26

The zero at $\Lambda = 1$ is now explained exactly by [Equation \(113\)](#); it is not merely an empirical absence. The observed total and parity occupancies remain substantially below the sharpened deterministic envelopes.

20.9 Uniform local ratios

For a dyadic base N , define

$$\mathfrak{R}(N) := \max_{\substack{N \leq M \leq 2N-H \\ H \in \{1, 2, 4, \dots, N\}}} \frac{\sum_{n=M}^{M+H-1} |S_n|^2}{HM^2}. \quad (222)$$

Table 11: Worst local ratios from the exact range $n \leq 4096$.

base N	$\mathfrak{R}(N)$	$\mathfrak{R}(N)/N^{0.10}$	maximizing start	maximizing H
128	0.176193	0.108459	243	1
256	0.145830	0.083757	309	1
512	0.183003	0.098069	547	1
1024	0.157837	0.078918	1032	1
2048	0.166479	0.077665	2544	1

The fitted log–log slope of $\mathfrak{R}(N)$ over these five dyadic scales is

$$\hat{\theta} = -0.00494746. \quad (223)$$

This range passes the preliminary scale-stability test $\hat{\theta} \leq 0.10$, but it is not long enough to determine the asymptotic law.

20.10 Further diagnostics

The next numerical stage must resolve the complete joint Gram structure, not merely produce additional diagonal energies. At each dyadic scale it should report:

1. both the theorem-predicted coefficient $\lambda_\sigma a_C$ and the exact orthogonal coefficient $\hat{\beta}_C$ for every cofactor block;
2. the full height-shell Gram matrix $G_{\ell m}$ and its spectrum;
3. cross-shell contributions grouped by cofactor and prime ranges;
4. exact reduced-square phase classes modulo $2r$ for all retained major arcs;
5. the resonant/nonresonant leakage created by replacing $P_{\text{res},M}$ with $P_{\text{res},M'}$ under scale descent;
6. the low-height and endpoint forcing vectors before and after projection;
7. the contribution of every indexed cell to the full signed quantity $\mathbf{1}^* G_M \mathbf{1}$.

The preregistered pass condition is no longer that each shell or localized cell be small. It is that the full recombined Gram form obey a scale-uniform bound after the explicit major-arc modes have been modeled with modulus $2r$. A computation may therefore pass even when individual shell diagonals grow, provided the corresponding off-diagonal cancellation is stable and quantitatively explained. Conversely, a small aggregate residual without a stable cross-shell and cross-curve mechanism is not evidence for the required uniform theorem.

21 Machine-checked formalization

Why this section is here. Formalization is used here as a mathematical audit. It forces every index, modulus, projection, sign, and quantifier to be stated explicitly. In particular, it exposed the difference between a global cubic average and the genuinely RH-equivalent uniform local estimate.

The accompanying Lean 4 project is available at

https://github.com/OVV0-Financial/RH_Lea.

It is pinned to mathlib v4.24.0 and currently imports thirty-nine theorem modules. The source audit rejects `sorry`, `admit`, newly introduced axioms, and hidden theorem substitutes; the full library is built with warnings treated as errors.

The formal development is organized by dependency rather than by the narrative order of this paper.

21.1 Arithmetic, phase, and geometry

The compiled arithmetic layer includes Möbius doubling, four-slot compression, deterministic prime-3 activation, the prime-square congruence modulo 24, and the two square classes modulo 40. The

phase layer uses the canonical modulus $2r$, proves the shift-by- r sign law, defines the reduced Gauss factor over $(\mathbb{Z}/2r\mathbb{Z})^\times$, and keeps the prime-3 cell-mask energy separate from the prime-3 quadratic resonance.

The geometry layer formalizes Fermat midpoint/half-gap coordinates, squared-complex recovery, cofactor parabolas, the $2ab$ displacement laws, conformality of the square map, and the two-sheeted square fibre with positive-branch injectivity.

21.2 Signed Gram and residual architecture

The Hilbert-space layer proves the exact height-shell Gram identity with every off-diagonal real inner product retained. It defines the true orthogonal coefficient and residual, proves orthogonality and the Pythagorean identity, and keeps that projection separate from the theorem-predicted coefficient used in the number-theoretic decomposition.

Subsequent modules define the scale-dependent resonant projection, resonant/nonresonant leakage, the weighted affine Lyapunov closure, the fully indexed actual residual, exact resonant cancellation for certified Möbius cofactor pairs, forcing estimates, and the full shell/cofactor/mode/row signed Gram recombination. No theorem replaces the joint signed form by separate positive shell bounds.

21.3 Certificates, actual-start identities, and localization

The finite-range checker verifies exact integer checkpoint data and has a proved soundness theorem. Acceptance of a certificate is deliberately separate from the mathematical realization that identifies the checked data with the analytic quantities in this paper.

The actual-start modules formalize

$$\text{actual}(M) = 2 \text{prediction}(M) + \text{residual}(M)$$

and the exact signed energy identity

$$\begin{aligned} \text{actualEnergy} &= 4 \text{predictionEnergy} + \text{residualEnergy} \\ &\quad + \text{signedPredictionResidualInteraction}. \end{aligned}$$

The sharp constant-4 inequality is obtained only when the signed interaction explicitly absorbs the residual budget.

The corrected localization layer repeats these definitions on every interval $[N, N + H)$. It introduces a separate local signed-interaction hypothesis, proves the exact windowed energy identity, and proves the sharp local frame inequality. A prefix estimate beginning at zero is never used as a substitute for a translated local estimate.

21.4 The concrete geometric–Mertens bridge

The formal RH target is the manuscript’s uniform local criterion, not merely the global benchmark $V(N) = O(N^{3+\epsilon})$. The modules `SquarePrefixMertensBridge`, `ConcreteSquarePrefixGeometry`, and `MathlibMertensHook` now define and connect the exact chain

$$\begin{aligned} \text{high-sector uniform local criterion} &\iff \text{total uniform local criterion} \iff \text{pointwise square-prefix criterion} \\ &\iff \text{exact square-prefix Mertens criterion} \iff \text{standard Mertens growth criterion}. \end{aligned}$$

The first equivalence is proved from exact signal recombination and the translated-window low-sector bound; no energy subtraction identity or cross-term estimate is used. The remaining project-side equivalences prove the exact endpoint $X_n = (n + 1)^2 - 1$, the interval Lipschitz estimate from $|\mu| \leq 1$, square-block interpolation, exponent conversion, and pointwise/local conversion.

The zero-friction theorem `squarePrefix_highUniformLocalBounded_iff_riemannHypothesis_of_classical_iff` accepts exactly one external theorem argument of type

$$\text{MertensEnergyBoundedStatement} \iff \text{RiemannHypothesisStatement}.$$

No abstract start sequence, realization equality, square-index adapter, exponent adapter, localization adapter, or project-specific RH bridge remains in that signature. Thus all reconciliation specific to this framework is complete. The classical Mertens theorem remains an explicit external dependency until it is available in mathlib, and the high-sector estimate itself remains the open analytic theorem. The formalization does not claim RH.

22 Fixed packets, small-prime channels, and uniform control

23 Fixed packets, small-prime channels, and uniform control

Why this section is here. The packets themselves do not change when the observation horizon grows. Making that fixedness explicit removes a false source of operator evolution and isolates the real scale-dependent leakage.

This section isolates a logical point that is easy to obscure when one works with a family of finite matrices indexed by the scale. The arithmetic packets themselves do not change as the horizon grows. What changes is only which fixed packets are being observed and how they are normalized. This distinction removes a spurious source of “operator evolution” and places the remaining proof obligation in its sharpest form.

23.1 A universal fixed packet system

Retain the complete half-square grid

$$X_m = \left\lfloor \frac{m^2}{2} \right\rfloor$$

and the complete block sums

$$\Delta_m := M(X_{m+1}) - M(X_m) = \sum_{X_m < n \leq X_{m+1}} \mu(n).$$

For every starting index u and length L , define the complete packet

$$\mathcal{B}_{u,L} := (X_u, X_{u+L}], \quad S_{u,L} := \sum_{m=u}^{u+L-1} \Delta_m. \quad (224)$$

Once u and L are fixed, both $\mathcal{B}_{u,L}$ and $S_{u,L}$ are fixed forever. Passing to a larger horizon does not alter any earlier block or packet.

Proposition 23.1 (No retroactive packet deformation). *For every u, L and every observation horizon $H \geq u + L$, the value of $S_{u,L}$ is independent of H . If a finite packet vector is enlarged by appending additional packet coordinates, its previously existing coordinates are unchanged.*

Proof. The value in Equation (224) is a finite sum of the already fixed block values $\Delta_u, \dots, \Delta_{u+L-1}$. No later block occurs in that sum. $\square$

It is therefore preferable to define one universal incidence system. For an arithmetic source w , let

$$\Phi_w(u, L) := \mathbf{1}_{\{w \in \mathcal{B}_{u,L}\}}.$$

The infinite column Φ_w is fixed. A scale- M operator is merely a coordinate restriction

$$P_M \Phi_w = (\Phi_w(u, M))_{M \leq u < 2M}.$$

If A_∞ denotes the universal synthesis map, then

$$A_M = P_M A_\infty, \quad K_M = A_\infty^* P_M^* P_M A_\infty. \quad (225)$$

Thus K_M changes with M only because the observation projection changes. The arithmetic columns do not move.

For nested square prefixes the same point is even more transparent. Let

$$p_n(w) := \mathbf{1}_{\{e(w) \leq n\}}.$$

Then the prefix kernel satisfies the exact rank-one update

$$K_{N+1} = K_N + p_{N+1} \otimes p_{N+1}, \quad (226)$$

and consequently

$$V(N+1) = V(N) + |S_{N+1}|^2. \quad (227)$$

No previous term is changed.

23.2 The role of the $2ab$ coordinate

The fixed-packet formulation corrects the interpretation of the squared height. The coordinate

$$Y = 2ab = \frac{q^2 - c^2}{2}$$

does not describe how an old packet deforms when a new packet is added. It determines the permanent activation interval of the canonical factor point. If

$$e(cq) = \lfloor \sqrt{cq} \rfloor, \quad h(cq) = q - 1,$$

then

$$I(c, q) := \{n : e(cq) \leq n \leq h(cq)\}$$

is fixed, and its incidence column is

$$\phi_{c,q}(n) = \mathbf{1}_{I(c,q)}(n).$$

The exact continuous lifetime identity from [Section 10](#) may be written

$$Y = \frac{q\tau}{2} \left(1 + \sqrt{c/q}\right) (1 + c/q), \quad \tau = q - \sqrt{cq}. \quad (228)$$

Hence Y/q controls the length of a fixed incidence interval. Differences in Y control differences among fixed columns; they do not cause retroactive changes.

For the low-frequency atoms

$$v_r(\theta) = \mathbf{1}_{[-1/N, 1/N]}(\theta) e(r\theta),$$

one has, for $|r - s| \ll N$,

$$\|v_r - v_s\|_2^2 = \frac{8\pi^2}{3} \frac{(r - s)^2}{N^3} + O\left(\frac{(r - s)^4}{N^5}\right). \quad (229)$$

Consequently, the $2ab$ spacing controls a modulus of continuity for the fixed packet columns. This is the functional-analytic image of the squared-space spacing law.

23.3 Least-prime channels

Every squarefree integer $n > 1$ has the unique small-prime representation

$$n = pa, \quad p = P^-(n), \quad P^-(a) > p, \quad \mu(pa) = -\mu(a).$$

For the fixed packet $\mathcal{B}_{u,M}$, define

$$L_{p,M}(u) := - \sum_{\substack{X_u/p < a \leq X_{u+M}/p \\ \mu(a) \neq 0, \ P^-(a) > p}} \mu(a). \quad (230)$$

Then, coordinatewise,

$$\mathbf{S}_M = \sum_p \mathbf{L}_{p,M}. \quad (231)$$

This is an identity among fixed packet vectors. The $p = 2$ channel is the exact sign-reversed odd squarefree parent process at numerical scale $1/2$, corresponding to block scale $M/\sqrt{2}$. Every $p \geq 3$ parent lies at numerical scale at most $1/3$.

23.4 Dyadic sign-flip cells

The prime-2 mechanism is stronger than a channel-level heuristic. It compresses the raw Möbius sequence into an exact permanent four-slot cell before averaging, projection, or asymptotic estimation.

Theorem 23.2 (Exact dyadic sign reversal). *For every odd integer $a \geq 1$,*

$$\boxed{\mu(2a) = -\mu(a)}. \quad (232)$$

This includes nonsquarefree a , for which both sides vanish.

Proof. If a is squarefree, then $\gcd(2, a) = 1$ and multiplicativity gives $\mu(2a) = \mu(2)\mu(a) = -\mu(a)$. If a is not squarefree, the same prime square divides both a and $2a$, so both Möbius values are zero. $\square$

Taking $a = 2k + 1$ and using $4 \mid 4k + 4$ yields

$$\mu(4k + 2) = -\mu(2k + 1), \quad \mu(4k + 4) = 0. \quad (233)$$

Theorem 23.3 (Exact four-slot cell compression). *Define*

$$\mathbf{C}_k := (\mu(4k + 1), \mu(4k + 2), \mu(4k + 3), \mu(4k + 4)).$$

Then

$$\boxed{\mathbf{C}_k = (\mu(4k + 1), -\mu(2k + 1), \mu(4k + 3), 0).} \quad (234)$$

Its scalar compression is

$$\boxed{C_k = \mu(4k + 1) - \mu(2k + 1) + \mu(4k + 3).} \quad (235)$$

The pattern $(+, -, +, 0)$ is therefore exact. At the fixed-column level,

$$\mu(a)\phi_a + \mu(2a)\phi_{2a} = \mu(a)(\phi_a - \phi_{2a}), \quad (236)$$

so the common component cancels before the packet norm is formed.

Universal prime-3 activation

Since $4 \equiv 1 \pmod{3}$, the three active slots have residues $k + 1, k + 2, k$ modulo 3.

Theorem 23.4 (Prime-3 active-slot cycle). *Exactly one of $4k + 1, 4k + 2, 4k + 3$ is divisible by 3. The affected slot is*

$k \bmod 3$	<i>active multiple of 3</i>
0	$4k + 3$
1	$4k + 2$
2	$4k + 1$

(237)

Thus prime 2 first creates the exact three-active-channel cell, and prime 3 then acts on one compulsory channel in every cell. Up to cyclic translation, the ideal live sign mask is

$$m_3 = (+1, -1, +1),$$

whose normalized Fourier squared magnitudes are

$$\frac{1}{9}, \quad \frac{4}{9}, \quad \frac{4}{9}. \quad (238)$$

The mean-mode energy factor $1/9$ concerns the cell-index mask only. By ?? 12.6, it does not suppress the exactly coherent quadratic prime phase at $\alpha = 1/3$.

Complete-cell and half-square packet equivalence

For a finite cell-index interval I ,

$$\boxed{\sum_{n \in 4I + \{1,2,3,4\}} \mu(n) = \sum_{k \in I} C_k.} \quad (239)$$

For an arbitrary integer interval $(A, B]$, complete cells leave only two endpoint fragments, so

$$\sum_{A < n \leq B} \mu(n) = \sum_{k \in \mathcal{K}(A,B)} C_k + E(A, B), \quad |E(A, B)| \leq 6. \quad (240)$$

For the half-square grid $X_m = \lfloor m^2/2 \rfloor$, each endpoint is 0 or 2 modulo 4. A fragment adjacent to an endpoint of class 2 contains a multiple of 4, whose Möbius value is zero. Therefore the fixed packet $\mathcal{B}_{u,M} = (X_u, X_{u+M}]$ satisfies

$$\boxed{S_{u,M} = \sum_{k=\lceil X_u/4 \rceil}^{\lfloor X_{u+M}/4 \rfloor - 1} [\mu(4k+1) - \mu(2k+1) + \mu(4k+3)] + E_{u,M},} \quad (241)$$

with

$$\boxed{|E_{u,M}| \leq 3, \quad \sum_{u=M}^{2M-1} |E_{u,M}|^2 \leq 9M.} \quad (242)$$

Define

$$\tilde{S}_{u,M} := \sum_{k=\lceil X_u/4 \rceil}^{\lfloor X_{u+M}/4 \rfloor - 1} [\mu(4k+1) - \mu(2k+1) + \mu(4k+3)]. \quad (243)$$

Then $S_{u,M} = \tilde{S}_{u,M} + E_{u,M}$ with boundary energy $O(M)$, so the raw and compressed packet-energy criteria are equivalent at the $M^{3+\varepsilon}$ scale. The primary actual-start problem is therefore

$$\boxed{\sum_{u=M}^{2M-1} |\tilde{S}_{u,M}|^2 \ll_{\varepsilon} M^{3+\varepsilon}.} \quad (244)$$

The open operator estimate acts on this already compressed signed process.

23.5 Small-prime cell contraction versus rational resonances

The prime-2 parity mask has the exact low-mode factor

$$\rho_2 = \frac{35 + 3\sqrt{65}}{70} = 0.845525332 \dots \quad (245)$$

For the ideal cell mask, the prime-3 mean-mode energy factor is

$$\rho_{3,\text{cell}} = \frac{1}{9}. \quad (246)$$

These factors can enter a global contraction budget only after the corrected rational major-arc component has been projected out or explicitly included in the parent state. The exact modes from [Corollary 12.3](#) are not leakage errors; they are genuine coherent modes requiring cofactor cancellation.

23.6 The signed Gram form

Separate the fixed child-feature columns by the sign of the parent cofactor and write

$$Z_+ = \sum_{\mu(a)=+1} \Phi_{a,M}, \quad Z_- = \sum_{\mu(a)=-1} \Phi_{a,M}.$$

The exact signed packet state is $Z_+ - Z_-$. If

$$A = \|Z_+\|^2, \quad B = \|Z_-\|^2, \quad \cos \vartheta = \frac{\operatorname{Re} \langle Z_+, Z_- \rangle}{\sqrt{AB}},$$

then

$$\|Z_+ - Z_-\|^2 = \left(\sqrt{A} - \sqrt{B}\right)^2 + 2\sqrt{AB}(1 - \cos \vartheta). \quad (247)$$

The two terms are respectively the directional mass defect and the directional profile defect. The required cancellation is therefore not generic orthogonality; it is near-concordance of the two opposite-sign aggregate profiles.

Equivalently, with synthesis operators U_M and V_M for the two sign classes, the joint Gram matrix is

$$\mathbb{G}_M = \begin{pmatrix} U_M^* U_M & U_M^* V_M \\ V_M^* U_M & V_M^* V_M \end{pmatrix} \succeq 0.$$

A natural sufficient condition is a small pair of Schur-complement defects, together with spectral balance of $U_M^* U_M$ and $V_M^* V_M$. This is the operator form of concordance saturation.

23.7 Uniform bounds and the quantifier issue

The fixedness of every finite packet does not by itself give the RH estimate. The necessary assertion is a uniform bound with one constant independent of the scale. Define

$$q_\varepsilon(N) := \frac{V(N)}{N^{3+\varepsilon}}, \quad Q_\varepsilon(N) := \sup_{1 \leq m \leq N} q_\varepsilon(m). \quad (248)$$

The target is

$$\boxed{\sup_{N \geq 1} q_\varepsilon(N) < \infty.} \quad (249)$$

This is the precise sense in which the proof requires a finite bound with no growing N -dependent coefficient. It is not enough that for each fixed N one may choose a new constant C_N .

Theorem 23.5 (Uniform closure from strict scale descent). *Suppose that there exist N_0 , $0 < \rho < 1$, $\eta > 0$, and $C_\varepsilon < \infty$, all independent of N , such that for $N \geq N_0$,*

$$q_\varepsilon(N) \leq \rho Q_\varepsilon\left(N/\sqrt{2} + O(1)\right) + C_\varepsilon N^{-\eta}. \quad (250)$$

Then Equation (249) holds. In particular,

$$V(N) \ll_\varepsilon N^{3+\varepsilon}.$$

Proof. Iterate Equation (250) until the ancestral scale is at most N_0 . After $k \asymp \log N$ steps,

$$q_\varepsilon(N) \leq \rho^k Q_\varepsilon(N_0) + C_\varepsilon \sum_{j=0}^{k-1} \rho^j \left(N 2^{-j/2} \right)^{-\eta}.$$

Choose $\eta > 0$ small enough that $\rho 2^{\eta/2} < 1$. The geometric series is uniformly bounded, and the first term tends to zero. Enlarging the final constant to cover $N < N_0$ proves the claim. $\square$

The theorem shows exactly why the fixed-packet insight is useful but not circular. It eliminates retroactive deformation and leaves only a uniform smaller-scale comparison. Once the coefficient ρ is strictly below one, the finite uniform bound is elementary.

23.8 The resonant/nonresonant channel budget

Let $P_{\text{res},M}$ be the scale-dependent rational-mode projection from Equation (172). The candidate prime budget applies only to the nonresonant diagonal action after all retained modulus- $2r$ modes have been extracted. Its orientation value remains

$$\rho_{\text{nr,cand}} = 2^{-3/2} \rho_2 + 3^{-3/2} \rho_{3,\text{cell}} + \rho_{\geq 5,\text{nr}}, \quad (251)$$

and, conditionally on

$$\rho_{\geq 5,\text{nr}} \leq \sum_{p \geq 5} p^{-3/2} + o(1), \quad (252)$$

the numerical orientation

$$2^{-3/2} \rho_2 + 3^{-3/2} \frac{1}{9} + \sum_{p \geq 5} p^{-3/2} \approx 0.624 \quad (253)$$

leaves a margin below one. This is not a bound for the full state, and it is not even sufficient for the projected state unless resonant/nonresonant leakage and additive forcing are controlled. The facts $|\gamma(1, 3)| = 1$ and $\rho_{3,\text{cell}} = 1/9$ remain compatible because they act in different Fourier variables.

To expose the feedback explicitly, let

$$R_M = P_{\text{res},M} Z_M, \quad N_M = P_{\text{nr},M} Z_M.$$

A parent-to-child descent has the schematic form

$$\begin{pmatrix} R_M \\ N_M \end{pmatrix} = \begin{pmatrix} A_M & B_M \\ C_M & D_M \end{pmatrix} \begin{pmatrix} R_{\text{par}(M)} \\ N_{\text{par}(M)} \end{pmatrix} + \begin{pmatrix} f_M^R \\ f_M^N \end{pmatrix}. \quad (254)$$

Here B_M and C_M include projection leakage, while f_M^R, f_M^N contain low-height, endpoint, and completion forcing. A bound on D_M alone does not close the argument.

Conjecture 23.6 (Joint actual-start block Lyapunov contraction). *For every $\varepsilon > 0$ there exist $\eta > 0$, $0 < \rho < 1$, $C_\varepsilon < \infty$, and a positive definite scale-uniform Lyapunov form $\mathcal{L}$ such that, for every actual-start compressed packet state,*

$$\left\| \begin{pmatrix} A_M & B_M \\ C_M & D_M \end{pmatrix} \right\|_{\mathcal{L}} \leq \rho < 1, \quad (255)$$

and

$$\left\| \begin{pmatrix} f_M^R \\ f_M^N \end{pmatrix} \right\|_{\mathcal{L}}^2 \leq C_\varepsilon M^{3-\eta+\varepsilon}. \quad (256)$$

The Lyapunov energy is the full actual-start signed Gram form: sums over height shells, cofactor ranges, prime modes, and resonant/nonresonant components remain inside the norm. All Fejer weights, cross-cell terms, cross-prime terms, boundary fragments, and ancestral completion terms are retained.

A slightly weaker sufficient alternative is a triangular theorem in which the resonant component is bounded independently, the leakage into the nonresonant component is summable, and the nonresonant diagonal block contracts uniformly. Either formulation must prevent low-height or boundary forcing from being reprojected coherently into a later resonant mode.

Combining [conjecture 23.6](#) with [Theorem 23.5](#) yields the finite normalized bound. The corrected final target is therefore

$$\boxed{\begin{array}{l} \text{joint resonant cancellation + cross-shell signed Gram control} \\ \text{+ nonresonant contraction + leakage/forcing control.} \end{array}} \quad (257)$$

24 What is proved, formalized, and still open

Why this section is here. There are three different claims to track: a result may be proved in the manuscript, formalized in Lean, or still required for the RH implication. Combining those categories into a single word such as “proved” would obscure the actual state of the program.

The open theorem is therefore specific. At every scale, the finite-dimensional resonant modes must cancel across Möbius-weighted cofactor curves; the complementary state must contract; the off-diagonal shell, curve, row, and mode interactions must remain inside one signed Gram form; and the leakage and forcing created by scale descent must be controlled uniformly on translated windows. The numerical experiments show why this formulation is necessary: the shell diagonals can be large while their signed recombination is small.

The formal project sharpens this boundary further. It has checked the algebraic identities, the corrected modulus, the projection distinctions, the joint-index bookkeeping, the local window identities, the concrete Mertens function and square endpoint, square interpolation, pointwise/local conversion, and the exact total/high criterion equivalence. No project-specific Mertens or RH reconciliation remains. What remains is the concrete analytic estimate for the high-sector signed Gram form, together with any realization data needed by the chosen operator proof.

25 Conclusion and next theorem

The main contribution of this paper is a change in organization. Square-prefix Möbius sums are not treated as an undifferentiated cancellation problem. They are decomposed into exact arithmetic channels, embedded in a squared-complex geometry, and recombined as one signed Gram problem.

Table 12: Mathematical and formal status of the program.

Layer	Mathematical status	Lean status
Square-root smooth–transport decomposition and channel telescoping	Proved	Core identities represented in the compiled architecture
Squared-complex recovery, cofactor parabolas, $2ab$ displacement, conformality, and square fibres	Proved	Formalized
Dyadic Möbius compression, prime-3 activation, and prime-square congruences	Proved	Formalized
Modulus- $2r$ phases, reduced Gauss factors, and small-modulus resonance	Proved	Formalized
Height-shell Gram identity, true orthogonal residual, and resonant/nonresonant decomposition	Proved as exact structural identities	Formalized
Finite certificate checker and abstract weighted block closure	Proved as interfaces and conditional closure theorems	Formalized
Actual-start prefix and local signed-frame inequalities	Conditional on explicit realization, contraction, and signed-absorption data	Formalized with all premises visible
Concrete $M(x)$, exact endpoint $X_n = (n + 1)^2 - 1$, and square interpolation	Proved	Formalized
Pointwise $\iff$ uniform local square-prefix criterion	Proved	Formalized
Exact total $\iff$ high-sector local criterion from the low-sector bound	Proved	Formalized
Direct hook for the classical Mertens criterion $\iff$ RH	Standard external theorem	Exact one-argument interface formalized
Concrete local high-sector signed-Gram estimate	Open	Statement and reduction formalized
Unconditional proof of RH	Not claimed	Not claimed

The square-root cutoff gives the exact identity

$$S_n = A_n - T_n.$$

The squared point

$$w_m = m + i \frac{q^2 - c^2}{2}$$

then records the source integer and its canonical factor imbalance simultaneously. Fixed cofactors lie on parabolas, the prefix and smoothness conditions become moving boundaries, and their intersection is the exact transition $q = R$. The movement from transport to smoothness is internal bookkeeping and cancels identically.

The ordinate $Y = (q^2 - c^2)/2$ does more than draw the picture. It controls factor gap, local occupancy, and transport lifetime. This closes the low-height sector at the cubic scale without invoking global sign cancellation. Conformality of the square map explains the two-power geometric dilution that makes that scale natural.

The high-height problem is different. The numerical data show that separate shell energies are not small; only their signed recombination is. The correct target must therefore retain the full off-diagonal Gram structure, distinguish exact resonant modes from the nonresonant state, and control projection leakage under scale descent. Fixed packets and exact small-prime identities reduce the ambiguity but do not eliminate the need for this operator estimate.

The RH connection is now a pure geometric equivalence:

$$\text{RH} \iff \sum_{n=N}^{N+H-1} |S_n^{\text{high}}|^2 \ll_{\varepsilon, \Lambda} H N^{2+\varepsilon}, \quad 1 \leq H \leq N.$$

The low sector is removed by an unconditional $O_\Lambda(HN^2)$ bound, and the two triangle inequalities transfer the local criterion in both directions. Taking $H = 1$ and interpolating between consecutive square endpoints connects the total criterion to the classical Mertens theorem. Lean formalizes every project-specific step in this chain and exposes the classical Mertens–RH equivalence as the single external theorem interface.

The next theorem is therefore precise and purely analytic: prove the uniform translated- window high-sector signed-Gram estimate. The geometric reduction itself is closed; a proof of that remaining estimate would satisfy the RH-equivalent criterion rather than merely supply another sufficient bound.

A Notation

B Reproducibility

The paper source, Lean formalization, and numerical reproducibility materials are maintained at

<https://github.com/OVVO-Financial/RH.Lean>.

The numerical tables in [Section 20](#) are reproduced from the repository’s `reproducibility` directory by

Symbol	Meaning
B_n	square block $[n^2, (n+1)^2)$
R_n	square-root cutoff $n+1$
X_n	complete-prefix endpoint $R_n^2 - 1$
S_n	$M(X_n)$
A_n	signed R_n -smooth mass
T_n	one-large-prime cofactor transport statistic
c	canonical cofactor $m/P^+(m)$
q	canonical largest prime $P^+(m)$
a, b	signed Fermat coordinates $(c+q)/2, (q-c)/2$
$w = x + iy$	squared point $cq + i(q^2 - c^2)/2$
$Y = y$	signed ordinate $2ab$
ρ	$ w = (c^2 + q^2)/2$
U, V	null coordinates c^2, q^2
γ_c	fixed cofactor parabola
Γ_R	moving smooth-transport cutoff parabola
$e(m)$	first complete-prefix index containing m
$h(m)$	smoothness-transition index $q - 1$
$L(m)$	discrete transport lifetime
$\tau(c, q)$	signed continuous delay $q - \sqrt{cq}$
K_N	cumulative square-prefix kernel
$P_{\text{res}, M}$	scale-dependent orthogonal major-arc projection
$G_{\ell m}(N)$	joint height-shell Gram matrix

```
python reproducibility/squared_space_reproducibility_v3.py \\  
--nmax 4096 \\  
--out reproducibility/squared_space_repro_v2
```

using Python, NumPy, and Numba. No network data, probabilistic sampling, or floating-point primality test is used. The release directory records the Python package versions and SHA-256 hashes of the script and generated outputs.

B.1 Deterministic algorithm

B.2 Deterministic algorithm

1. Set

$$X_{\max} = (N_{\max} + 1)^2 - 1.$$

2. Run a linear sieve through $X_{\max}$ to compute:

$$\mu(m), \quad \text{spf}(m), \quad P^+(m), \quad \pi(m).$$

3. Form

$$S_n = M((n + 1)^2 - 1)$$

by integer prefix summation.

4. Construct A_n and T_n independently from the source entry/transition intervals:

$$e(m) = \lfloor \sqrt{m} \rfloor, \quad h(m) = P^+(m) - 1.$$

A squarefree source contributes to T on $[e, h)$ and to A on $[\max\{e, h\}, N_{\max}]$.

5. For every squarefree $m > 1$, put

$$q = P^+(m), \quad c = m/q, \quad 2Y = q^2 - c^2.$$

Verify the norm, curve, and lifetime identities using integers only.

6. Compute the cofactor-1 curve exactly as

$$S_n^{(1)} = -\pi(X_n),$$

then set

$$S_n^{(\geq 2)} = S_n - S_n^{(1)}$$

and evaluate all three terms in [Equation \(198\)](#).

7. Bin every squarefree source by the least dyadic threshold $C = 2^j$ containing its canonical cofactor $c = m/P^+(m)$. Cumulatively sum the bins to obtain $R_C(X_n)$ and $V_C(N)$ for every dyadic C and every reported N .

8. Evaluate the total, odd, and even low-height counts and incidences for $\Lambda = 1, 2, 4, 8, 16$.

9. Run the exact major-arc regression tests:

$$q^2 \equiv 1 \pmod{24} \quad (2000 < q \leq 4000),$$

verify the eight full-coherence frequencies in [Table 3](#), verify the old modulus-3 expression is zero, and verify the corrected normalized modulus-6 factor has unit norm.

10. Evaluate $\Re(N)$ for dyadic bases 128, 256, 512, 1024, 2048 and all dyadic window lengths $H \leq N$.

B.3 Generated output files

The script writes:

Table 13: Machine-readable reproducibility outputs.

File	Contents
run_summary.txt	range, source counts, exact violation counts, and local-ratio slope
prefix_energy.csv	S_N , total energy, and A - T component energies
prime_curve_decomposition.csv	$c = 1$, $c \geq 2$, cross term, correlation, and residual ratio
cofactor_cascade.csv	$R_C(X_N)$ and $V_C(N)$ for every dyadic C and reported N
low_height_diagnostics.csv	sharpened total/parity bounds and observed clustering/lifetime maxima
local_ratio_diagnostics.csv	worst local ratios and maximizing intervals
major_arc_regression.csv	exact mod-24 checks, rational test frequencies, and old-versus-corrected Gauss factors

For the reported run, the summary must read:

```

N_max: 4096
X_max: 16785408
prime_count: 1078362
squarefree_sources_m_ge_2: 10204280
max_observed_canonical_cofactor: 692835
max_dyadic_cofactor_threshold: 16777216
max_abs_S_minus_A_plus_T: 0
nonzero_prefix_identity_errors: 0
norm_identity_violations: 0
cofactor_curve_identity_violations: 0
height_lifetime_violations: 0
prime_square_mod24_violations_2000_4000: 0
old_modulus_3_factor_abs: 0
corrected_modulus_6_factor_abs: 1
cofactor_cascade_full_error: 0
local_ratio_log_log_slope: -0.004947461592

```

B.4 Extended $N = 16384$ diagnostic run

The extended values in [Section 20.7](#) were produced by a separate memory-limited computation reaching $X_N \approx 2.68 \times 10^8$. They are presented as numerical diagnostics, not as machine-checked theorems. A release-grade archive should include the exact source, compiler and package versions, all integer checkpoints, the shell Gram matrix, and cryptographic hashes. Until those files are attached, only the algebraic congruence statements and identities proved elsewhere in the paper have unconditional theorem status.

B.5 Integrity hashes

The SHA-256 hashes of the released script and machine-readable outputs are recorded in `reproducibility/SHA256SUMS.txt`. This avoids copying stale hashes into the manuscript when the numerical protocol is extended. The separate $N = 16384$ diagnostic archive remains explicitly distinguished from the fully scripted $N = 4096$ suite until its source and complete outputs are committed under the same protocol.

The exact checks are binary: any nonzero violation invalidates the run. The asymptotic diagnostics are not proofs; their purpose is to identify which geometric or analytic component fails first as the range is extended.

References

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